

INVERSE TRIGONOMETRIC FUNCTIONS

(A) GENERAL INTRODUCTION :

$\sin^{-1} x$, $\cos^{-1} x$, $\tan^{-1} x$ etc. denote angles or real numbers whose sine is x , whose cosine is x and whose tangent is x , provided that the answers given are numerically smallest available. These are also written as $\text{arc sin } x$, $\text{arc cos } x$ etc.

If there are two angles one positive & the other negative having same numerical value, then positive angle should be taken.

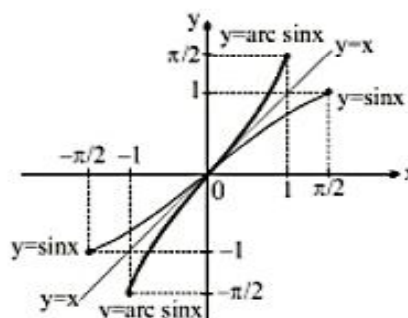
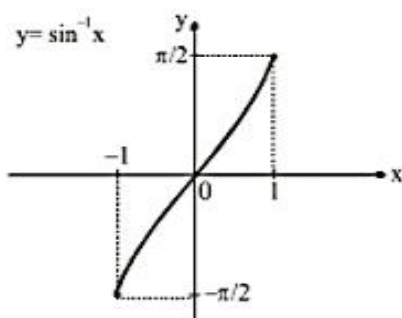
(B) PRINCIPAL VALUES AND DOMAINS OF INVERSE CIRCULAR FUNCTIONS :

S.No.	Function	Domain	Range
(i)	$y = \sin^{-1} x$	$-1 \leq x \leq 1$	$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
(ii)	$y = \cos^{-1} x$	$-1 \leq x \leq 1$	$0 \leq y \leq \pi$
(iii)	$y = \tan^{-1} x$	$x \in \mathbb{R}$	$-\frac{\pi}{2} < y < \frac{\pi}{2}$
(iv)	$y = \cot^{-1} x$	$x \in \mathbb{R}$	$0 < y < \pi$
(v)	$y = \text{cosec}^{-1} x$	$x \leq -1$ or $x \geq 1$	$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}, y \neq 0$
(vi)	$y = \sec^{-1} x$	$x \leq -1$ or $x \geq 1$	$0 \leq y \leq \pi; y \neq \frac{\pi}{2}$

- NOTE THAT :**
- (a) 1st quadrant is common to all the inverse functions.
 - (b) 3rd quadrant is **not** used in inverse functions.
 - (c) 4th quadrant is used in the **CLOCKWISE DIRECTION** i.e. $-\frac{\pi}{2} \leq y \leq 0$.

Graphs of all 6 inverse circular functions :

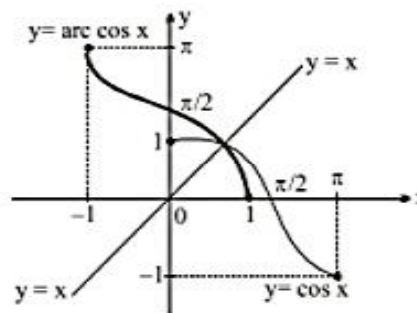
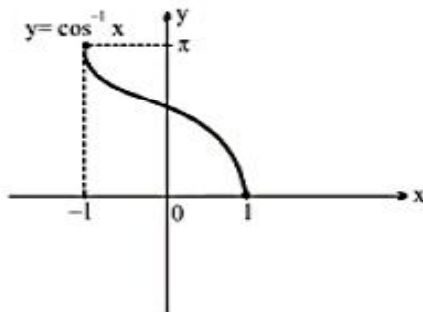
(I) $y = \sin^{-1} x, |x| \leq 1, y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$



Note : Graph of $y = \sin^{-1} x$ and $y = \sin x$ are mirror image of each other about the line $y = x$.

Highlights : -

- (i) $\sin^{-1}x$ is bounded in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.
 - (ii) $\sin^{-1}x$ is an odd function. (symmetric about origin)
 - (iii) $\sin^{-1}x$ is an increasing function in its domain.
 - (iv) Maximum value of $\sin^{-1}x = \frac{\pi}{2}$, occurs at $x = 1$ and minimum value of $\sin^{-1}x = -\frac{\pi}{2}$, occurs at $x = -1$.
 - (v) $\sin^{-1}x$ is an aperiodic function.
- (2) $y = \cos^{-1}x, |x| \leq 1, y \in [0, \pi]$

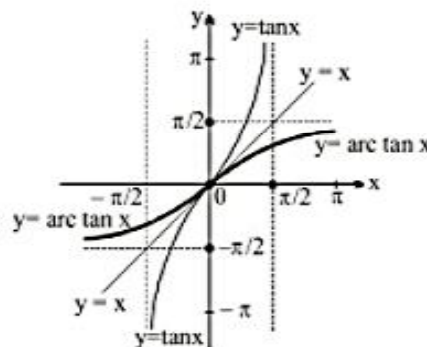
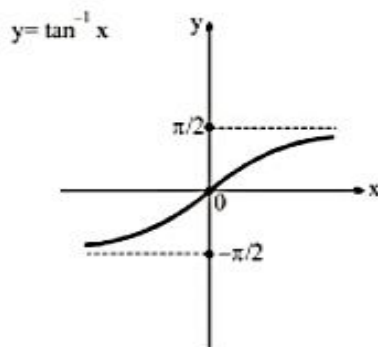


Note : Graph of $y = \cos^{-1}x$ and $y = \cos x$ are mirror image of each other about the line $y = x$.

Highlights : -

- (i) $\cos^{-1}x$ is bounded in $[0, \pi]$.
- (ii) $\cos^{-1}x$ is a neither odd nor even function.
- (iii) $\cos^{-1}x$ is a decreasing function in its domain.
- (iv) Maximum value of $\cos^{-1}x = \pi$, occurs at $x = -1$ and minimum value of $\cos^{-1}x = 0$, occurs at $x = 1$.
- (v) $\cos^{-1}x$ is an aperiodic function.

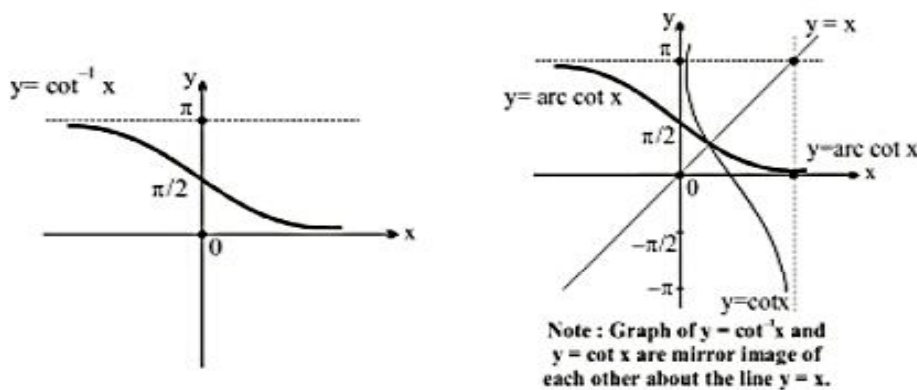
(3) $y = \tan^{-1}x, x \in \mathbb{R}, y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$



Note : Graph of $y = \tan^{-1}x$ and $y = \tan x$ are mirror image of each other about the line $y = x$.

Highlights : -

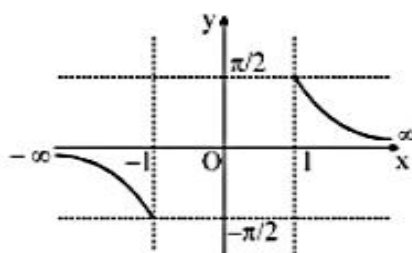
- (i) $\tan^{-1}x$ is bounded in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
 - (ii) $\tan^{-1}x$ is an odd function. (symmetric about origin)
 - (iii) $\tan^{-1}x$ is an increasing function in its domain.
 - (iv) $\tan^{-1}x$ is an aperiodic function.
- (4) $y = \cot^{-1}x, x \in \mathbb{R}, y \in (0, \pi)$



Highlights : -

- (i) $\cot^{-1}x$ is bounded in $(0, \pi)$.
- (ii) $\cot^{-1}x$ is a neither odd nor even function.
- (iii) $\cot^{-1}x$ is a decreasing function in its domain.
- (iv) $\cot^{-1}x$ is an aperiodic function.

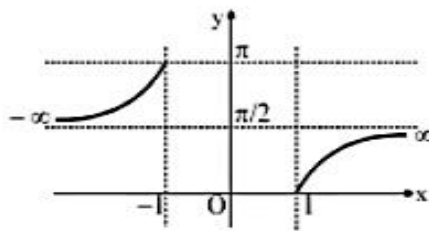
(5) $y = \operatorname{cosec}^{-1}x, |x| \geq 1, y \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$



Highlights : -

- (i) $\operatorname{cosec}^{-1}x$ is bounded in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.
- (ii) $\operatorname{cosec}^{-1}x$ is an odd function. (symmetric about origin)
- (iii) Maximum value of $\operatorname{cosec}^{-1}x = \frac{\pi}{2}$, occurs at $x = 1$ and minimum value of $\operatorname{cosec}^{-1}x = -\frac{\pi}{2}$, occurs at $x = -1$.
- (iv) $\operatorname{cosec}^{-1}x$ is a decreasing function.
- (v) $\operatorname{cosec}^{-1}x$ is an aperiodic function.

$$(6) \quad y = \sec^{-1} x, |x| \geq 1, \quad y \in \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$$



Highlights : -

- (i) $\sec^{-1} x$ is bounded in $[0, \pi]$.
- (ii) $\sec^{-1} x$ is a neither odd nor even function.
- (iii) Maximum value of $\sec^{-1} x = \pi$, occurs at $x = -1$ and minimum value of $\sec^{-1} x = 0$, occurs at $x = 1$.
- (iv) $\sec^{-1} x$ is an increasing function.
- (v) $\sec^{-1} x$ is an aperiodic function.

Note :

- (a) $\tan^{-1}(x)$ and $\cot^{-1}(x)$ are continuous and monotonic on $\mathbb{R} \Rightarrow$ that their range is $\mathbb{R}$
- (b) If $f(x)$ is continuous and has a range $\mathbb{R} \Rightarrow$ it is monotonic. e.g. $y = x^3 - 3x$.

Illustration :

Find domain and range of the following

(a) $\sin^{-1}[x]$

(b) $\cos^{-1}\{x\}$

(c) $\sin^{-1}(e^x)$

(d) $f(x) = \tan^{-1}(\log_{4/5}(5x^2 - 8x + 4))$

(where $[x]$ denotes the greatest integer function and $\{x\}$ denotes the fractional part function.)

Sol.

(a) $\sin^{-1}[x]$ defined when

$$-1 \leq [x] \leq 1 \Rightarrow -1 \leq x < 2$$

domain : $x \in [-1, 2)$

In this domain $[x]$ takes the values $-1, 0, 1$

$$\Rightarrow \text{range of } \sin^{-1}[x] = \{\sin^{-1} -1, \sin^{-1} 0, \sin^{-1} 1\}$$

$$\text{Range} = \left\{-\frac{\pi}{2}, 0, \frac{\pi}{2}\right\}$$

(b) $\cos^{-1}\{x\}$ defined when

$$-1 \leq \{x\} \leq 1$$

$$\Rightarrow \text{domain : } x \in \mathbb{R} \quad (\because \{x\} \in [0, 1))$$

$$\text{Range} = \cos^{-1}[0, 1)$$

$$= (\cos^{-1} 1, \cos^{-1} 0]$$

$$\text{Range} = \left(0, \frac{\pi}{2}\right]$$

(c) $\sin^{-1} e^x$ defined when
 $-1 \leq e^x \leq 1 \Rightarrow e^x \geq -1$ holds always true
 So $e^x \leq 1 \Rightarrow x \leq 0$
 domain $x \in (-\infty, 0]$
 In this domain $e^x \in (0, 1]$
 $\Rightarrow$ Range of $\sin^{-1} e^x = \sin^{-1}(0, 1]$
 $= \sin^{-1}(0, 1]$
 $= (\sin^{-1} 0, \sin^{-1} 1]$
 $\text{Range} = \left(0, \frac{\pi}{2}\right]$

(d) $f(x)$ is defined when $5x^2 - 8x + 4 > 0$
 $\because a > 0, D < 0 \Rightarrow 5x^2 - 8x + 4 > 0$ is true for all $x \in R$
 $\Rightarrow$ domain : $x \in R$
 Now, $5x^2 - 8x + 4 = 5 \left[\left(x - \frac{4}{5}\right)^2 + \frac{4}{25} \right]$
 $\Rightarrow 5x^2 - 8x + 4 \in \left[\frac{4}{5}, \infty\right)$ for $x \in R$
 $\Rightarrow$ Range of $f(x) = \tan^{-1} \left(\log_{4/5} \left[\frac{4}{5}, \infty \right) \right)$
 $\text{Range} = \tan^{-1} (-\infty, 1] = \left(-\frac{\pi}{2}, \frac{\pi}{4}\right]$

Illustration :

Find the value of

(a) $\sin \left(2 \sin^{-1} \frac{3}{5} \right)$

(b) $\cos (2 \tan^{-1} 2) + \sin (2 \tan^{-1} 3)$

(c) $\cos \left(\arcsin \frac{4}{5} - \arccos \frac{4}{5} \right)$

(d) $\tan \left(2 \cot^{-1} 5 - \frac{\pi}{4} \right)$

Sol.

(a) $\sin^{-1} \frac{3}{5} = \theta \Rightarrow \sin \theta = \frac{3}{5}, \cos \theta = \frac{4}{5}$

$\sin (2\theta) = 2 \sin \theta \cdot \cos \theta$

$= 2 \left(\frac{3}{5} \right) \left(\frac{4}{5} \right) = \frac{24}{25}$

$\left[\text{Ans. } \frac{24}{25} \right]$

(b) Let $\tan^{-1} 2 = \theta \Rightarrow \tan \theta = 2$
 $\tan^{-1} 3 = \phi \Rightarrow \tan \phi = 3$

Now

$$\begin{aligned}\cos(2\theta) + \sin(2\phi) &= \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} + \frac{2 \tan \phi}{1 + \tan^2 \phi} \\ &= \frac{1 - (2)^2}{1 + (2)^2} + \frac{2(3)}{1 + (3)^2} \\ &= \frac{-3}{5} + \frac{3}{5} = 0 \quad [Ans. 0]\end{aligned}$$

(c) Let $\sin^{-1} \frac{4}{5} = \theta \Rightarrow \sin \theta = \frac{4}{5}, \cos \theta = \frac{3}{5}$

$\cos^{-1} \frac{4}{5} = \phi \Rightarrow \cos \phi = \frac{4}{5}, \sin \phi = \frac{3}{5}$

Now

$$\begin{aligned}\cos(\theta - \phi) &= \cos \theta \cos \phi + \sin \theta \sin \phi \\ &= \frac{4}{5} \cdot \frac{3}{5} + \frac{4}{5} \cdot \frac{3}{5} = \frac{24}{25} \quad [Ans. \frac{24}{25}]\end{aligned}$$

(d) let $\cot^{-1} 5 = \theta \Rightarrow \cot \theta = 5, \tan \theta = \frac{1}{5}$

Now

$$\begin{aligned}\tan\left(2\theta - \frac{\pi}{4}\right) &= \frac{\tan 2\theta - 1}{1 + \tan 2\theta} \\ &= \frac{\left(\frac{2 \tan \theta}{1 - \tan^2 \theta}\right) - 1}{1 + \left(\frac{2 \tan \theta}{1 - \tan^2 \theta}\right)} = \frac{\left(\frac{2\left(\frac{1}{5}\right)}{1 - \left(\frac{1}{5}\right)^2}\right) - 1}{1 + \left(\frac{2\left(\frac{1}{5}\right)}{1 - \left(\frac{1}{5}\right)^2}\right)} = \frac{\frac{5}{12} - 1}{1 + \frac{5}{12}} = \frac{-7}{17} \quad [Ans. \frac{-7}{17}]\end{aligned}$$

Illustration :

Find the domain of definition of following functions.

(a) $f(x) = \arccos \frac{2x}{1+x}$

(b) $f(x) = \sin^{-1}\left(\frac{x-3}{2}\right) - \log_{10}(4-x)$

(c) $f(x) = \sqrt{3-x} + \cos^{-1}\left(\frac{3-2x}{5}\right) + \log_6(2|x| - 3) + \sin^{-1}(\log_2 x)$

Sol.

$$(a) \quad f(x) = \cos^{-1} \frac{2x}{1+x}$$

$$f(x) \text{ defined when } -1 \leq \frac{2x}{1+x} \leq 1$$

$$\text{Now } \frac{2x}{1+x} \geq -1 \Rightarrow \frac{2x}{1+x} + 1 \geq 0 \Rightarrow \frac{3x+1}{1+x} \geq 0$$

$$\Rightarrow x \in (-\infty, -1) \cup \left[-\frac{1}{3}, \infty\right) \quad \dots (i)$$

$$\text{and } \frac{2x}{1+x} \leq 1 \Rightarrow \frac{2x}{1+x} - 1 \leq 0 \Rightarrow \frac{x-1}{1+x} \leq 0$$

$$\Rightarrow x \in (-1, 1] \quad \dots (ii)$$

from (i) and (ii)

$$x \in \left[-\frac{1}{3}, 1\right]$$

$$\text{Ans. } x \in \left[-\frac{1}{3}, 1\right]$$

$$(b) \quad f(x) = \sin^{-1}\left(\frac{x-3}{2}\right) - \log_{10}(4-x)$$

$$f(x) \text{ defined when } -1 \leq \frac{x-3}{2} \leq 1 \text{ and } 4-x > 0$$

$$\text{Now } -1 \leq \frac{x-3}{2} \leq 1$$

$$\Rightarrow -2 \leq x-3 \leq 2$$

$$\Rightarrow 1 \leq x \leq 5$$

$$\Rightarrow x \in [1, 5] \quad \dots (i)$$

$$\text{and } 4-x > 0$$

$$\Rightarrow x < 4 \Rightarrow x \in (-\infty, 4) \quad \dots (ii)$$

$$\Rightarrow \text{from (i) and (ii)}$$

$$x \in [1, 4)$$

$$\text{Ans. } x \in [1, 4)$$

$$(c) \quad f(x) = \sqrt{3-x} + \cos^{-1}\left(\frac{3-2x}{5}\right) + \log_6(2|x|-3) + \sin^{-1}(\log_2 x)$$

$$\text{Now } 3-x \geq 0 \Rightarrow x \leq 3 \Rightarrow x \in (-\infty, 3] \quad \dots (i)$$

$$-1 \leq \frac{3-2x}{5} \leq 1 \Rightarrow -5 \leq 3-2x \leq 5$$

$$\Rightarrow -2 \leq 2x \leq 8 \Rightarrow -1 \leq x \leq 4$$

$$\Rightarrow x \in [-1, 4] \quad \dots (ii)$$

$$2|x| - 3 > 0 \Rightarrow |x| > \frac{3}{2} \Rightarrow x \in \left(-\infty, -\frac{3}{2}\right) \cup \left(\frac{3}{2}, \infty\right) \quad \dots (iii)$$

$$-1 \leq \log_2 x \leq 1 \Rightarrow \frac{1}{2} \leq x \leq 2 \Rightarrow x \in \left[\frac{1}{2}, 2\right] \quad \dots (iv)$$

from (i), (ii), (iii) and (iv)

$$x \in \left[\frac{3}{2}, 2\right] \quad \text{Ans. } \left[\frac{3}{2}, 2\right]$$

Illustration :

If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi$, then the value of $x^{2012} + y^{2012} + z^{2012} + \frac{6}{x^{2011} + y^{2011} + z^{2011}}$ is

equal to

- (A) 0 (B) 1 (C) -1 (D) 2

Sol. $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi$ is possible only when $\cos^{-1} x = \cos^{-1} y = \cos^{-1} z = \pi$
(all take their maximum value)

$$\Rightarrow x = -1, y = -1, z = -1$$

Now

$$\begin{aligned} & x^{2012} + y^{2012} + z^{2012} + \frac{6}{x^{2011} + y^{2011} + z^{2011}} \\ &= (-1)^{2012} + (-1)^{2012} + (-1)^{2012} + \frac{6}{(-1)^{2011} + (-1)^{2011} + (-1)^{2011}} \\ &= 3 + \frac{6}{-3} = 3 - 2 = 1 \quad \text{Ans.} \end{aligned}$$

Illustration :

Equation of the image of the line $x + y = \sin^{-1}(a^6 + 1) + \cos^{-1}(a^4 + 1) - \tan^{-1}(a^2 + 1)$, $a \in \mathbb{R}$ about x axis is given by

- (A) $x - y = 0$ (B) $x - y = \frac{\pi}{2}$ (C) $x - y = \pi$ (D) $x - y = \frac{\pi}{4}$

Sol. $\because \sin^{-1}$ is defined for $[-1, 1]$
 $\therefore a = 0$

$$\therefore x + y = \sin^{-1} 1 + \cos^{-1} 1 - \tan^{-1} 1 = \frac{\pi}{4}$$

$$\text{Clearly image about } x \text{ axis will be } x - y = \frac{\pi}{4} \quad \text{Ans.}$$

Illustration :

If $\sin^{-1} \left(x^2 - \frac{x^4}{3} + \frac{x^6}{9} - \dots \right) + \cos^{-1} \left(x^4 - \frac{x^8}{3} + \frac{x^{12}}{9} - \dots \right) = \frac{\pi}{2}$, where $0 \leq |x| < \sqrt{3}$, then

number of values of 'x' is equal to

- (A) 1 (B) 2 (C) 3 (D) 4

Sol. $\sin^{-1} \underbrace{\left(x^2 - \frac{x^4}{3} + \frac{x^6}{9} - \dots \right)}_X + \cos^{-1} \underbrace{\left(x^4 - \frac{x^8}{3} + \frac{x^{12}}{9} - \dots \right)}_Y = \frac{\pi}{2}$

Now $X = Y$

$$\frac{x^2}{1 + \frac{x^2}{3}} = \frac{x^4}{1 + \frac{x^4}{3}} \Rightarrow \frac{3}{3+x^2} = \frac{3x^2}{3+x^4} \Rightarrow 9 + 3x^4 = 9x^2 + 3x^4 \Rightarrow x^2 = 1$$

$$\Rightarrow x = 0, 1 \text{ or } -1$$

$\therefore$ Number of values is equal to 3. **Ans.**

Illustration :

If $0 < \cos^{-1} x < 1$ and $1 + \cos^{-1} x + (\cos^{-1} x)^2 + \dots \infty = 2$ then x is equal to

- (A) $\frac{\pi}{4}$ (B) $\cos \frac{1}{2}$ (C) $\cos \frac{1}{\sqrt{2}}$ (D) $\frac{\pi}{6}$

Sol. We have

$1 + \cos^{-1} x + (\cos^{-1} x)^2 + \dots \infty = 2$, (which is an infinite geometric progression)

$$\Rightarrow \frac{1}{1 - \cos^{-1} x} = 2 \Rightarrow \cos^{-1} x = \frac{1}{2} \Rightarrow x = \cos \frac{1}{2} \text{ Ans.}$$

Illustration :

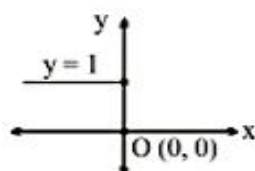
Let $f(x) = \frac{|\cos^{-1}(\operatorname{sgn} x)|}{\cos^{-1}(\operatorname{sgn} x)}$ then which of the following is (are) not correct?

[Note: $\operatorname{sgn} x$ denotes signum function of x .]

- (A) Range of $f(x)$ contains no integer.
(B) Graph of $f(x)$ is symmetric about y-axis.
(C) The equation $f(x) = 0$ has two distinct real solutions.
(D) Inverse of $f(x)$ is not defined.

Sol. Clearly, $D_f = (-\infty, 0]$

Now, $f(x) = 1 \forall x \in (-\infty, 0]$ **Ans.**



Practice Problem

Q.1 Find domain and range of the following

(a) $\cos^{-1}[x]$ (b) $\sin^{-1}\{x\}$ (c) $\cot^{-1}(\operatorname{sgn} x)$ (d) $\cot^{-1} \log_{\frac{4}{5}}(5x^2 - 8x + 4)$

(where $[x]$ denotes the greatest integer function and $\{x\}$ denotes the fractional part function.)

Q.2 Find the value of

(a) $\tan\left(\frac{1}{2}\cos^{-1}\frac{\sqrt{5}}{3}\right)$ (b) $\sin(\tan^{-1}2) + \cos(\tan^{-1}2)$

Q.3 State which of the statements are True or False ?

- (i) $y = \operatorname{sgn}(\cot^{-1}x)$ and $y = 1$ identical. (ii) $e^{\ln(\tan^{-1}x)}$ and $\tan^{-1}x$ identical.
(iii) $e^{\ln(\cot^{-1}x)}$ and $\cot^{-1}x$ identical.

Q.4 Find domain of definition the functions $f(x) = \sqrt{\cos(\sin x)} + \sin^{-1} \frac{1+x^2}{2x}$

Q.5 The range of the function $y = \left(\frac{\cos^{-1}(3x-1)}{\pi} + 1\right)^2$ is

- (A) $[1,4]$ (B) $[0,\pi]$ (C) $[1,\pi]$ (D) $[0,\pi^2]$

Q.6 If $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \frac{3\pi}{2}$, then possible value(s) of a satisfying the equation

$$x^{100} + y^{100} + z^{100} + \frac{a^2}{x^{50} + y^{50} + z^{50}} = \frac{10a}{3} \text{ are}$$

- (A) 1 (B) 4 (C) 9 (D) 16

Answer key

Q.1 (a) $D : x \in [-1, 2)$ and $R \in \left\{0, \frac{\pi}{2}, \pi\right\}$ (b) $D : x \in \mathbb{R}; R : [0, \pi/2)$

(c) $D : x \in \mathbb{R}; R : \left\{\frac{\pi}{2}, \frac{\pi}{4}, \frac{3\pi}{4}\right\}$ (d) $D : x \in \mathbb{R}; R : \left[\frac{\pi}{4}, \pi\right)$

Q.2 (a) $\frac{3-\sqrt{5}}{2}$, (b) $\frac{3}{\sqrt{5}}$

Q.3 (i) True ; (ii) False ; (iii) True

Q.4 $\{-1, 1\}$

Q.5 A

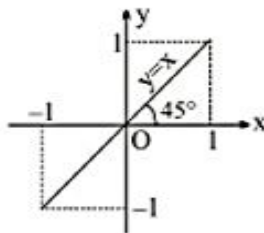
Q.6 A,C

PROPERTIES OF INVERSE TRIGONOMETRIC FUNCTION :

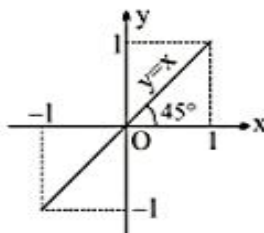
Property-1 :

- | | |
|---|--|
| (i) $\sin(\sin^{-1} x) = x, -1 \leq x \leq 1$ | (ii) $\cos(\cos^{-1} x) = x, -1 \leq x \leq 1$ |
| (iii) $\tan(\tan^{-1} x) = x, x \in \mathbb{R}$ | (iv) $\cot(\cot^{-1} x) = x, x \in \mathbb{R}$ |
| (v) $\operatorname{cosec}(\operatorname{cosec}^{-1} x) = x, x \geq 1$ | (vi) $\sec(\sec^{-1} x) = x, x \geq 1$ |

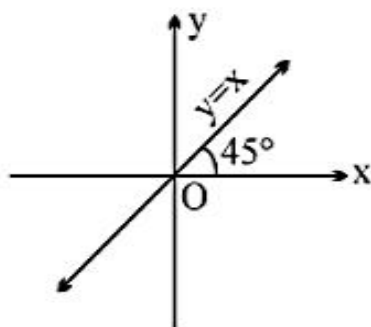
- (1) $y = \sin(\sin^{-1} x) = x, x \in [-1, 1], y \in [-1, 1], y$ is aperiodic.



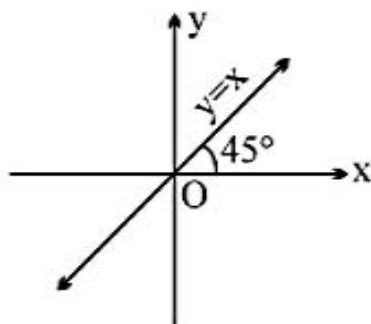
- (2) $y = \cos(\cos^{-1} x) = x, x \in [-1, 1], y \in [-1, 1], y$ is aperiodic.



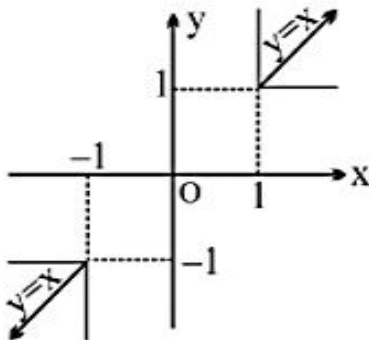
- (3) $y = \tan(\tan^{-1} x) = x, x \in \mathbb{R}, y \in \mathbb{R}, y$ is aperiodic.



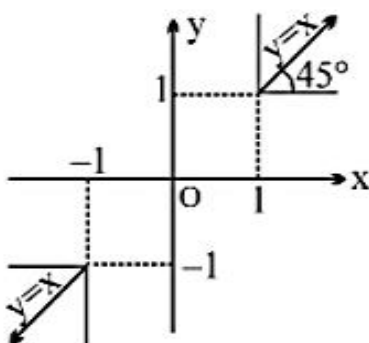
- (4) $y = \cot(\cot^{-1} x) = x, x \in \mathbb{R}, y \in \mathbb{R}, y$ is aperiodic.



- (5) $y = \operatorname{cosec}(\operatorname{cosec}^{-1}x) = x, |x| \geq 1, |y| \geq 1$, y is aperiodic.



- (6) $y = \sec(\sec^{-1}x) = x, |x| \geq 1, |y| \geq 1$, y is aperiodic.



Note that: (1, 2); (3, 4) and (5, 6) are identical function.

(vii) $\sin^{-1}(\sin x) = x, -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ (vii) $\cos^{-1}(\cos x) = x; 0 \leq x \leq \pi$

(ix) $\tan^{-1}(\tan x) = x; -\frac{\pi}{2} < x < \frac{\pi}{2}$ (x) $\cot^{-1}(\cot x) = x, 0 < x < \pi$

(xi) $\operatorname{cosec}^{-1}(\operatorname{cosec} x) = x; -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}, x \neq 0$

(xii) $\sec^{-1}(\sec x) = x; 0 \leq x \leq \pi, x \neq \frac{\pi}{2}$

- (7) $y = \sin^{-1}(\sin x), x \in \mathbb{R}, y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, Periodic with period 2π

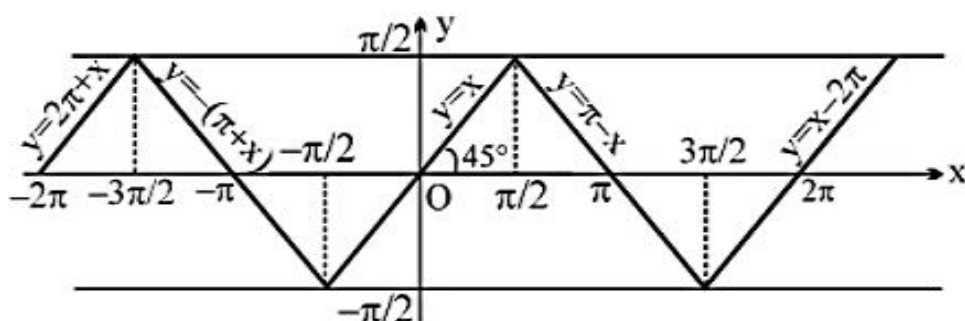


Illustration :

Find the value of following

- (a) $\sin^{-1}(\sin 1)$ (b) $\sin^{-1}(\sin 2)$ (c) $\sin^{-1}(\sin 3)$
(d) $\sin^{-1}(\sin 4)$ (e) $\sin^{-1}(\sin 5)$ (f) $\sin^{-1}(\sin 10)$

Sol. (a) $\sin^{-1}(\sin 1) = 1$ ($\because 1 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$)

(b) $\sin^{-1}(\sin 2) \neq 2$ ($\because 2 \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$)

$\Rightarrow \sin^{-1}(\sin 2) = \sin^{-1}(\sin(\pi - 2)) = \pi - 2$ ($\because \pi - 2 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$)

(c) $\sin^{-1}(\sin 3) = \sin^{-1}(\sin(\pi - 3)) = \pi - 3$ ($\because \pi - 3 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$)

(d) $\sin^{-1}(\sin 4) = \sin^{-1}(\sin(\pi - 4)) = \pi - 4$ ($\because \pi - 4 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$)

(e) $\sin^{-1}(\sin 5) = \sin^{-1}(\sin(5 - 2\pi)) = 5 - 2\pi$ ($\because 5 - 2\pi \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$)

(f) $\sin^{-1}(\sin 10) = \sin^{-1}(\sin(3\pi - 10)) = 3\pi - 10$ ($\because 3\pi - 10 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$)

(8) $y = \cos^{-1}(\cos x) = x, x \in \mathbb{R}, y \in [0, \pi]$, periodic with period 2π

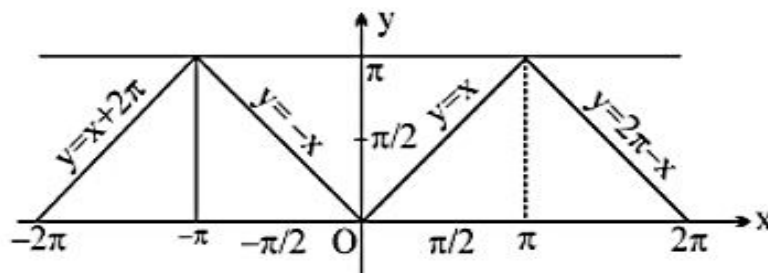


Illustration :

Find the value of following

- (a) $\cos^{-1}(\cos 1)$ (b) $\cos^{-1}(\cos 2)$ (c) $\cos^{-1}(\cos 3)$
(d) $\cos^{-1}(\cos 4)$ (e) $\cos^{-1}(\cos 5)$ (f) $\cos^{-1}(\cos 10)$

Sol. (a) $\cos^{-1}(\cos 1) = 1$; ($\because 1 \in [0, \pi]$)

(b) $\cos^{-1}(\cos 2) = 2$; ($\because 2 \in [0, \pi]$)

(c) $\cos^{-1}(\cos 3) = 3$; ($\because 3 \in [0, \pi]$)

(d) $\cos^{-1}(\cos 4) = \cos^{-1}(\cos(2\pi - 4)) = 2\pi - 4$; ($\because 2\pi - 4 \in [0, \pi]$)

(e) $\cos^{-1}(\cos 5) = \cos^{-1}(\cos(2\pi - 5)) = 2\pi - 5$; ($\because 2\pi - 5 \in [0, \pi]$)

(f) $\cos^{-1}(\cos 10) = \cos^{-1}(\cos(10 - 3\pi)) = 10 - 3\pi$ ($\because 10 - 3\pi \in [0, \pi]$)

- (9) $y = \tan^{-1}(\tan x) = x, x \in \mathbb{R} - \left\{ (2n-1)\frac{\pi}{2}, n \in \mathbb{I} \right\}, y \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right),$
periodic with period π

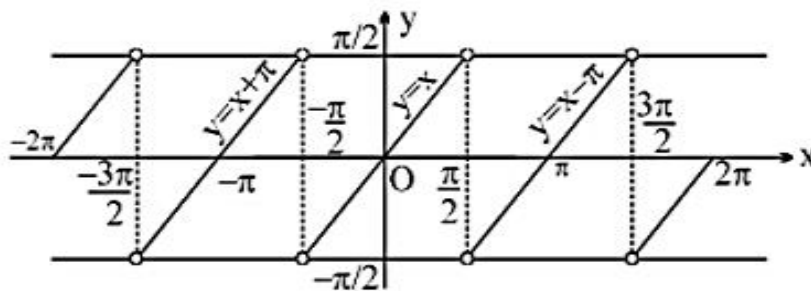


Illustration :

Find the value of following

(a) $\tan^{-1}(\tan 1)$

(b) $\tan^{-1}(\tan 2)$

(c) $\tan^{-1}(\tan 3)$

(d) $\tan^{-1}(\tan 4)$

(e) $\tan^{-1}(\tan 5)$

(f) $\tan^{-1}(\tan 10)$

Sol. (a) $\tan^{-1}(\tan 1) = 1 \quad (\because 1 \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right))$

(b) $\tan^{-1}(\tan 2) \neq 2 \quad (\because 2 \notin \left(-\frac{\pi}{2}, \frac{\pi}{2} \right))$

$\Rightarrow \tan^{-1}(\tan 2) = \tan^{-1}(\tan(\pi - 2)) = \pi - 2 \quad (\because \pi - 2 \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right))$

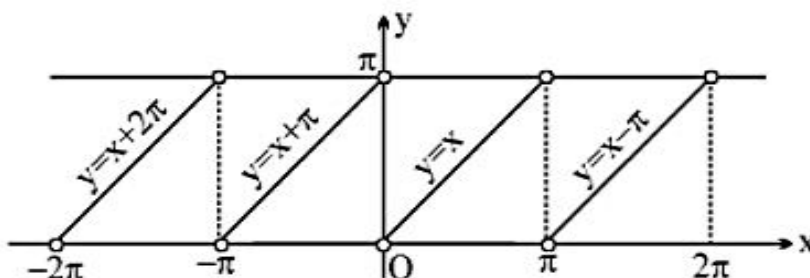
(c) $\tan^{-1}(\tan 3) = \tan^{-1}(\tan(\pi - 3)) = \pi - 3 \quad (\because \pi - 3 \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right))$

(d) $\tan^{-1}(\tan 4) = \tan^{-1}(\tan(\pi - 4)) = \pi - 4 \quad (\because \pi - 4 \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right))$

(e) $\tan^{-1}(\tan 5) = \tan^{-1}(\tan(5 - 2\pi)) = 5 - 2\pi \quad (\because 5 - 2\pi \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right))$

(f) $\tan^{-1}(\tan 10) = \tan^{-1}(\tan(3\pi - 10)) = 3\pi - 10 \quad (\because 3\pi - 10 \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right))$

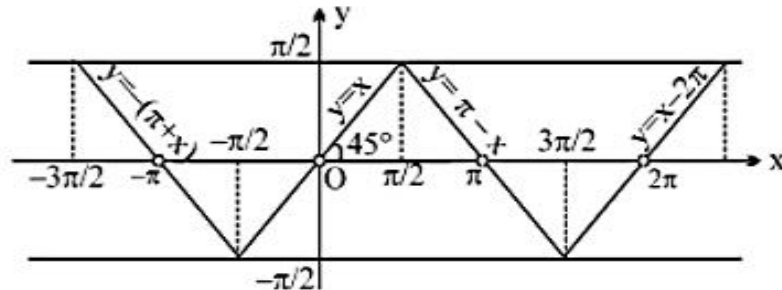
- (10) $y = \cot^{-1}(\cot x) = x, x \in \mathbb{R} - \{n\pi\}, y \in (0, \pi),$ periodic with π



(11) $y = \operatorname{cosec}^{-1}(\operatorname{cosec} x)$, $x \in \mathbb{R} - \{n\pi, n \in \mathbb{I}\}$,

$$y \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$$

y is periodic with period 2π



(12) $y = \sec^{-1}(\sec x) = x$, y is periodic ;

$$x \in \mathbb{R} - \left\{(2n-1)\frac{\pi}{2} \mid n \in \mathbb{I}\right\}, y \in \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$$

with period 2π

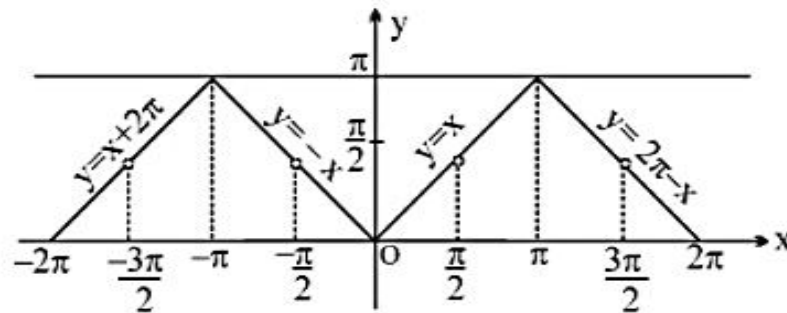


Illustration :

Find the integral solution of inequality $6x^2 - 5x < \cos^{-1}(\cos 5) - 2 \sin^{-1}(\sin 3)$.

Sol. $6x^2 - 5x < \cos^{-1}(\cos(2\pi - 5)) - 2 \sin^{-1}(\sin(\pi - 3))$

$$6x^2 - 5x < 2\pi - 5 - 2\pi + 6$$

$$6x^2 - 5x < 1$$

$$6x^2 - 5x - 1 < 0$$

$$(6x + 1)(x - 1) < 0$$

$$\Rightarrow x \in \left(-\frac{1}{6}, 1\right)$$

Integral solution is $x = 0$

Ans. $\{0\}$

Illustration :

If $\sin^{-1}(\sin 9) - \cos^{-1}(\cos 15)$ can be written in the form $a\pi - b$, then find the value of $a + b$. ($a, b \in \mathbb{N}$).

Sol. $\sin^{-1}(\sin 9) = \sin^{-1} \sin(3\pi - 9) = 3\pi - 9$ $(\because 3\pi - 9 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right])$

$\cos^{-1}(\cos 15) = \cos^{-1}(\cos(15 - 4\pi)) = 15 - 4\pi$ $(\because 15 - 4\pi \in [0, \pi])$

$$\begin{aligned}\Rightarrow \sin^{-1}(\sin 9) - \cos^{-1}(\cos 15) &= (3\pi - 9) - (15 - 4\pi) \\ &= 7\pi - 24 \Rightarrow a = 7, b = 24 \\ a + b &= 7 + 24 = 31.\end{aligned}$$

Ans. 31

Illustration :

Find the value of following

$$(a) \sin^{-1} \sin \left(\frac{13\pi}{11} \right) \quad (b) \cos^{-1} \left(\sin \left(-\frac{\pi}{4} \right) \right) \quad (c) \sin^{-1} \left(\cos \frac{33\pi}{10} \right)$$

Sol. (a) $\sin^{-1} \sin \left(\frac{13\pi}{11} \right) = \sin^{-1} \sin \left(\pi + \frac{2\pi}{11} \right) = \sin^{-1} \left(-\sin \left(\frac{2\pi}{11} \right) \right) = \sin^{-1} \sin \left(-\frac{2\pi}{11} \right) = -\frac{2\pi}{11} \text{ Ans.}$

(b) $\cos^{-1} \sin \left(-\frac{\pi}{4} \right) = \cos^{-1} \cos \left(\frac{\pi}{2} + \frac{\pi}{4} \right) = \cos^{-1} \cos \left(\frac{3\pi}{4} \right) = \frac{3\pi}{4} \text{ Ans.}$

(c) $\sin^{-1} \cos \left(\frac{33\pi}{10} \right) = \sin^{-1} \cos \frac{13\pi}{10} = \sin^{-1} \left(-\cos \frac{3\pi}{10} \right) = \sin^{-1} \left(-\sin \left(\frac{5\pi}{10} - \frac{3\pi}{10} \right) \right)$
 $= \sin^{-1} \left(-\sin \frac{\pi}{5} \right) = \sin^{-1} \left(\sin \left(-\frac{\pi}{5} \right) \right) = -\frac{\pi}{5} \text{ Ans.}$

Illustration :

If $\sin^{-1} \left(\sin \frac{33\pi}{7} \right) + \cos^{-1} \left(\cos \frac{46\pi}{7} \right) + \tan^{-1} \left(-\tan \frac{13\pi}{8} \right) + \cot^{-1} \left(\cot \left(-\frac{19\pi}{8} \right) \right)$

can be written in the form of $\frac{a\pi}{b}$ (where $a, b \in \mathbb{N}$) then find the minimum value of $a + b$.

Sol. $\sin^{-1} \left(\sin \left(\frac{33\pi}{7} \right) \right) = \sin^{-1} \left(\sin \left(5\pi - \frac{2\pi}{7} \right) \right) = \sin^{-1} \left(\sin \left(\frac{2\pi}{7} \right) \right) = \frac{2\pi}{7}$

$\cos^{-1} \left(\cos \left(\frac{46\pi}{7} \right) \right) = \cos^{-1} \left(\cos \left(6\pi + \frac{4\pi}{7} \right) \right) = \cos^{-1} \left(\cos \left(\frac{4\pi}{7} \right) \right) = \frac{4\pi}{7}$

$\tan^{-1} \left(-\tan \left(\frac{13\pi}{8} \right) \right) = \tan^{-1} \left(\tan \left(2\pi - \frac{13\pi}{8} \right) \right) = \tan^{-1} \left(\tan \left(\frac{3\pi}{8} \right) \right) = \frac{3\pi}{8}$

$\cot^{-1} \left(\cot \left(-\frac{19\pi}{8} \right) \right) = \cot^{-1} \left(\cot \left(3\pi - \frac{19\pi}{8} \right) \right) = \cot^{-1} \left(\cot \left(\frac{5\pi}{8} \right) \right) = \frac{5\pi}{8}$

$\Rightarrow \sin^{-1} \left(\sin \frac{33\pi}{7} \right) + \cos^{-1} \left(\cos \frac{46\pi}{7} \right) + \tan^{-1} \left(-\tan \frac{13\pi}{8} \right) + \cot^{-1} \left(\cot \left(-\frac{19\pi}{8} \right) \right)$

$= \frac{2\pi}{7} + \frac{4\pi}{7} + \frac{3\pi}{8} + \frac{5\pi}{8} = \frac{6\pi}{7} + \pi = \frac{13\pi}{7}$

$\Rightarrow a = 13, b = 7 \Rightarrow a + b = 13 + 7 = 20$

Ans. 20

Illustration :

The smallest positive integral value of n for which

$$(n-2)x^2 + 8x + n + 4 > \sin^{-1}(\sin 12) + \cos^{-1}(\cos 12) \quad \forall x \in R, \text{ is}$$

(A) 4 (B) 5 (C) 6 (D) 7

Sol. We have

$$\sin^{-1}(\sin 12) + \cos^{-1}(\cos 12) = -(4\pi - 12) + (4\pi - 12) = 0$$

$$\therefore (n-2)x^2 + 8x + n + 4 > 0 \quad \forall x \in R$$

$$\Rightarrow (n-2) > 0 \Rightarrow n \geq 3 \text{ and } (8)^2 - 4(n-2)(n+4) < 0 \text{ or } n^2 + 2n - 24 > 0$$

$$\Rightarrow n > 4 \Rightarrow n \geq 5$$

$$\text{So, } n_{\text{smallest}} = 5. \text{ Ans.}$$

Illustration :

The product of all real values of x satisfying the equation

$$\sin^{-1} \cos \left(\frac{2x^2 + 10|x| + 4}{x^2 + 5|x| + 3} \right) = \cot \left(\cot^{-1} \left(\frac{2 - 18|x|}{9|x|} \right) \right) + \frac{\pi}{2} \text{ is}$$

(A) 9 (B) -9 (C) -3 (D) -1

Sol.
$$\frac{\pi}{2} - \cos^{-1} \cos \left(\frac{2(x^2 + 5|x| + 3) - 2}{x^2 + 5|x| + 3} \right) = \cot \cot^{-1} \left(\frac{2}{9|x|} - 2 \right) + \frac{\pi}{2}$$

$$\frac{\pi}{2} - 2 + \frac{2}{x^2 + 5|x| + 3} = \frac{2}{9|x|} - 2 + \frac{\pi}{2}$$

$$\Rightarrow |x|^2 - 4|x| + 3 = 0$$

$$|x| = 1, 3 \Rightarrow x = \pm 1, \pm 3$$

$$\Rightarrow \text{Product} = (1)(-1)(3)(-3) = 9 \text{ Ans.}$$

Illustration :

Which of the following is/are correct?

(A) $\cos(\cos(\cos^{-1} 1)) < \sin(\sin^{-1}(\sin(\pi - 1))) < \sin(\cos^{-1}(\cos(2\pi - 2)))$

(B) $\cos(\cos(\cos^{-1} 1)) < \sin(\cos^{-1}(\cos(2\pi - 2))) < \sin(\sin^{-1}(\sin(\pi - 1))) < \tan(\cot^{-1}(\cot 1))$

(C) $\sum_{t=1}^{5000} \cos^{-1}(\cos(2t\pi - 1)) = \sum_{t=1}^{2500} \cot^{-1}(\cot(t\pi + 2))$ where $t \in I$

(D) $\cot^{-1} \cot \operatorname{cosec}^{-1} \operatorname{cosec} \sec^{-1} \sec \tan \tan^{-1} \cos \cos^{-1} \sin^{-1} \sin 4 = 4 - \pi$

Sol. for (A) and (B)

$$\cos(\cos^{-1} 1) = 1 \Rightarrow \cos(\cos(\cos^{-1} 1)) = \cos 1$$

$$\sin^{-1}(\sin(\pi - 1)) = \pi - (\pi - 1) = 1 \Rightarrow \sin(\sin^{-1}(\sin(\pi - 1))) = \sin 1$$

$$\cos^{-1}(\cos(2\pi - 2)) = \cos^{-1}(\cos 2) = 2 \Rightarrow \sin(\cos^{-1}(\cos(2\pi - 2))) = \sin 2$$

$$\tan(\cot^{-1}(\cot 1)) = \tan 1$$

It is easy to compare $\cos 1$, $\sin 1$, $\sin 2$, $\tan 1$

$$\cos 1 < \sin 1 < \sin 2 < \tan 1 \Rightarrow (A) \text{ is correct}$$

for (C)

$\cos^{-1} \cos x$ is periodic and even

$$\cos^{-1} \cos(2t\pi - 1) = \cos^{-1}(\cos 1) = 1 \quad (t \in I)$$

$$\sum_{t=1}^{5000} \cos^{-1} \cos(2t\pi - 1) = 5000$$

now $\cot^{-1} \cot(t\pi + 2) = 2$ [$\cot^{-1} \cot x$ is periodic with period π]

$$\therefore \sum_{t=1}^{2500} \cot^{-1} \cot(t\pi + 2) = 5000 \Rightarrow (C) \text{ is correct}$$

$$(D) \quad \sin^{-1} \sin 4 = \pi - 4$$

$$\cos \cos^{-1}(\pi - 4) = \pi - 4$$

$$\tan \tan^{-1}(4 - \pi) = \pi - 4$$

$$\sec^{-1} \sec(\pi - 4) = 4 - \pi$$

$$\operatorname{cosec}^{-1} \operatorname{cosec}(4 - \pi) = 4 - \pi$$

$$\cot^{-1} \cot(4 - \pi) = 4 - \pi \Rightarrow (D) \text{ is correct}$$

Practice Problem

Q.1 Find the value of following

$$(i) \sin^{-1}[\cos 2 \cot^{-1}(\sqrt{2} - 1)]$$

$$(ii) \sin^{-1}(\sin 7) + \cos^{-1} \cos(13)$$

$$(iii) \sin^{-1}\left(\sin \frac{10\pi}{7}\right)$$

$$(iv) \cos^{-1}\left(\sin\left(-\frac{\pi}{9}\right)\right)$$

Q.2 If $3 \leq a < 4$ then the value of $\sin^{-1}(\sin [a]) + \tan^{-1}(\tan [a]) + \sec^{-1}(\sec [a])$, where $[x]$ denotes greatest integer function less than or equal to x , is equal to

$$(A) 3$$

$$(B) 2\pi - 9$$

$$(C) 2\pi - 3$$

$$(D) 9 - 2\pi$$

Q.3 The value of $\sin^{-1}(\cos 2) - \cos^{-1}(\sin 2) + \tan^{-1}(\cot 4) - \cot^{-1}(\tan 4) + \sec^{-1}(\operatorname{cosec} 6) - \operatorname{cosec}^{-1}(\sec 6)$ is

$$(A) 0$$

$$(B) 3\pi$$

$$(C) 8 - 3\pi$$

$$(D) 5\pi - 16$$

Paragraph for question nos. 4 to 6

$$\text{For } x \in \left(0, \frac{\pi}{4}\right),$$

$$\text{Let } S_n = \sum_{r=1}^{2n} \sin(\sin^{-1} x^{3r-2}), C_n = \sum_{r=1}^{2n} \cos(\cos^{-1} x^{3r-1}) \text{ and } T_n = \sum_{r=1}^{2n} \tan(\tan^{-1} x^{3r})$$

where $n \in \mathbb{N}$ and $n \geq 3$.

Q.4 The correct order of S_n , C_n and T_n is given by

$$(A) S_n > T_n > C_n$$

$$(B) S_n < C_n < T_n$$

$$(C) S_n < T_n < C_n$$

$$(D) S_n > C_n > T_n$$

Q.5 The value of $\lim_{n \rightarrow \infty} (S_n + C_n + T_n)$ is equal to

- (A) $\frac{1}{1-x}$ (B) $\frac{x}{1-x}$ (C) $\frac{1}{1+x}$ (D) $\frac{x}{1+x}$

Q.6 The value of 'x' for which $S_n = C_n + T_n$, is

- (A) $\sin \frac{\pi}{5}$ (B) $2 \sin \frac{\pi}{5}$ (C) $2 \sin \frac{\pi}{10}$ (D) $\sin \frac{\pi}{10}$

Answer key

Q.1 (i) $-\frac{\pi}{4}$; (ii) $20 - 6\pi$; (iii) $-\frac{3\pi}{7}$; (iv) $\frac{11\pi}{18}$

Q.2 A Q.3 D Q.4 D Q.5 B Q.6 C

Property-2 :

- (1) $\operatorname{cosec}^{-1}x = \sin^{-1} \frac{1}{x} ; |x| \geq 1$
- (2) $\sin^{-1}x = \operatorname{cosec}^{-1} \frac{1}{x}, |x| \leq 1, x \neq 0$
- (3) $\sec^{-1}x = \cos^{-1} \frac{1}{x} ; |x| \geq 1$
- (4) $\cos^{-1}x = \sec^{-1} \frac{1}{x}, |x| \leq 1, x \neq 0$
- (5) $\cot^{-1}x = \tan^{-1} \frac{1}{x} ; x > 0$
 $= \pi + \tan^{-1} \frac{1}{x} ; x < 0$

Note : (i) $\operatorname{cosec}^{-1}x$ and $\sin^{-1} \frac{1}{x}$ are identical function.

(ii) $\sin^{-1}x$ and $\operatorname{cosec}^{-1} \frac{1}{x}$ are not identical because domain of $\sin^{-1}x$ and $\operatorname{cosec}^{-1} \frac{1}{x}$ is not equal.

(iii) $\sec^{-1}x$ and $\cos^{-1} \frac{1}{x}$ are identical function.

(iv) $\cos^{-1}x$ and $\sec^{-1} \frac{1}{x}$ are not identical because domain of $\cos^{-1}x$ and $\sec^{-1} \frac{1}{x}$ is not equal.

Illustration :

Are $\tan(\cot^{-1}x)$ and $\cot(\tan^{-1}x)$ are identical ?

Sol. [True], as both functions have same graph.

Illustration :

Find the value(s) of x satisfying the equation

$$\cot^{-1} \frac{x^2-1}{2x} + \tan^{-1} \frac{2x}{x^2-1} = \frac{2\pi}{3}$$

Sol. Case-(i) $\frac{x^2-1}{2x} > 0$

$$\cot^{-1} \frac{x^2-1}{2x} + \tan^{-1} \frac{2x}{x^2-1} = \frac{2\pi}{3} \quad (\because \cot^{-1} x = \tan^{-1} \frac{1}{x}; x > 0)$$

$$\Rightarrow \tan^{-1} \frac{2x}{x^2-1} + \tan^{-1} \frac{2x}{x^2-1} = \frac{2\pi}{3} \Rightarrow \tan^{-1} \frac{2x}{x^2-1} = \frac{\pi}{3}$$

$$\Rightarrow \frac{2x}{x^2-1} = \sqrt{3} \Rightarrow \sqrt{3}x^2 - 2x - \sqrt{3} = 0 \Rightarrow x = \frac{2 \pm 4}{2\sqrt{3}} \Rightarrow x = \frac{-1}{\sqrt{3}}, \sqrt{3}$$

Case-(ii)

$$\frac{x^2-1}{2x} < 0$$

$$\cot^{-1} \frac{x^2-1}{2x} + \tan^{-1} \frac{2x}{x^2-1} = \frac{2\pi}{3}$$

$$\Rightarrow \pi + \tan^{-1} \frac{2x}{x^2-1} + \tan^{-1} \frac{2x}{x^2-1} = \frac{2\pi}{3} \quad (\because \cot^{-1} x = \pi + \tan^{-1} \frac{1}{x}; x < 0)$$

$$\Rightarrow \tan^{-1} \frac{2x}{x^2-1} = \frac{-\pi}{6} \Rightarrow \frac{2x}{x^2-1} = \frac{-1}{\sqrt{3}} \Rightarrow x^2 + 2\sqrt{3}x - 1 = 0$$

$$\Rightarrow x = -\sqrt{3} \pm 2 \Rightarrow x = -(2 + \sqrt{3}), 2 - \sqrt{3}$$

From case (i) and (ii)

$$\Rightarrow x = \sqrt{3}, -\frac{1}{\sqrt{3}}, -(2 + \sqrt{3}), (2 - \sqrt{3}) \quad \text{Ans.}$$

Property-3 :

- (i) $\sin^{-1}(-x) = -\sin^{-1}x$, $-1 \leq x \leq 1$
- (ii) $\tan^{-1}(-x) = -\tan^{-1}x$, $x \in \mathbb{R}$
- (iii) $\cos^{-1}(-x) = \pi - \cos^{-1}x$, $-1 \leq x \leq 1$
- (iv) $\cot^{-1}(-x) = \pi - \cot^{-1}x$, $x \in \mathbb{R}$
- (v) $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1}x$, $|x| \geq 1$
- (vi) $\sec^{-1}(-x) = \pi - \sec^{-1}x$, $|x| \geq 1$

Property-4 :

- (i) $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$, $-1 \leq x \leq 1$
- (ii) $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$, $x \in \mathbb{R}$
- (iii) $\operatorname{cosec}^{-1}x + \sec^{-1}x = \frac{\pi}{2}$, $|x| \geq 1$

Illustration :

Find the value of x if

$$(a) \ 4\sin^{-1}x + \cos^{-1}x = \frac{3\pi}{4}; \quad (b) \ 5\tan^{-1}x + 3\cot^{-1}x = \frac{7\pi}{4}$$

Sol.

$$(a) \quad 4\sin^{-1}x + \frac{\pi}{2} - \sin^{-1}x = \frac{3\pi}{4}$$

$$\Rightarrow 3\sin^{-1}x = \frac{\pi}{4} \quad \Rightarrow \sin^{-1}x = \frac{\pi}{12} \quad \Rightarrow x = \sin \frac{\pi}{12} \quad \Rightarrow x = \frac{\sqrt{3}-1}{2\sqrt{2}} \text{ Ans.}$$

$$(b) \quad 5\tan^{-1}x + 3\left(\frac{\pi}{2} - \tan^{-1}x\right) = \frac{7\pi}{4}$$

$$2\tan^{-1}x = \frac{7\pi}{4} - \frac{3\pi}{2} \Rightarrow 2\tan^{-1}x = \frac{\pi}{4} \Rightarrow \tan^{-1}x = \frac{\pi}{8}$$

$$\Rightarrow x = \tan \frac{\pi}{8} \Rightarrow x = \sqrt{2}-1 \text{ Ans.}$$

Illustration :

Find the maximum and minimum values of $(\sin^{-1}x)^3 + (\cos^{-1}x)^3$

Sol. $(\sin^{-1}x)^3 + (\cos^{-1}x)^3 = (\sin^{-1}x + \cos^{-1}x)((\sin^{-1}x)^2 + (\cos^{-1}x)^2 - \sin^{-1}x \cdot \cos^{-1}x)$

$$= \frac{\pi}{2}((\sin^{-1}x) + (\cos^{-1}x))^2 - 3\sin^{-1}x \cdot \cos^{-1}x$$

$$= \frac{\pi}{2}\left[\left(\frac{\pi}{2}\right)^2 - 3\sin^{-1}x\left(\frac{\pi}{2} - \sin^{-1}x\right)\right] = \frac{\pi}{2}\left[\frac{\pi^2}{4} - \frac{3\pi}{2}\sin^{-1}x + 3(\sin^{-1}x)^2\right]$$

$$= \frac{3\pi}{2} \left[(\sin^{-1} x)^2 - \frac{\pi}{2} \sin^{-1} x + \frac{\pi^2}{12} \right] = \frac{3\pi}{2} \left[\left(\sin^{-1} x - \frac{\pi}{4} \right)^2 + \frac{\pi^2}{48} \right]$$

Maximum value occurs when $\sin^{-1} x = -\frac{\pi}{4}$

$$\Rightarrow \text{Maximum value} = \frac{3\pi}{2} \left[\left(-\frac{\pi}{4} - \frac{\pi}{4} \right)^2 + \frac{\pi^2}{48} \right] = \frac{3\pi}{2} \cdot \frac{7\pi^2}{12} = \frac{7\pi^3}{8}$$

Minimum value occurs when $\sin^{-1} x = \frac{\pi}{4}$

$$\Rightarrow \text{Minimum value} = \frac{3\pi}{2} \cdot \left[\frac{\pi^2}{48} \right] = \frac{\pi^3}{32}$$

Illustration :

Find the range of $f(x) = \sin^{-1} x + \cos^{-1} x + \tan^{-1} x$.

Sol. $f(x) = \sin^{-1} x + \cos^{-1} x + \tan^{-1} x$
domain : $x \in [-1, 1]$

$$\Rightarrow f(x) = \frac{\pi}{2} + [\tan^{-1} -1, \tan^{-1} 1] = \frac{\pi}{2} + \left[-\frac{\pi}{4}, \frac{\pi}{4} \right] = \left[\frac{\pi}{4}, \frac{3\pi}{4} \right]$$

Property-5 :

$$(1) \quad \tan^{-1} x + \tan^{-1} y = \begin{cases} \tan^{-1} \left(\frac{x+y}{1-xy} \right), & \text{if } xy < 1 \\ \pi + \tan^{-1} \left(\frac{x+y}{1-xy} \right), & \text{if } x > 0, y > 0 \text{ and } xy > 1 \\ -\pi + \tan^{-1} \left(\frac{x+y}{1-xy} \right), & \text{if } x < 0, y < 0 \text{ and } xy > 1 \end{cases}$$

Proof:

Let $\tan^{-1} x = A$ and $\tan^{-1} y = B$, where $A, B \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$.

$$\text{Now, } \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{x+y}{1-xy}$$

$$\Rightarrow \tan^{-1} \left(\frac{x+y}{1-xy} \right) = \tan^{-1} \tan(A+B)$$

$$= \tan^{-1} \tan \alpha, \text{ where } \alpha \in (-\pi, \pi)$$

$$\tan^{-1}\left(\frac{x+y}{1-xy}\right) = \tan^{-1}(\tan \alpha)$$

$$= \begin{cases} \alpha + \pi, & -\pi < \alpha < -\frac{\pi}{2} \\ \alpha, & -\frac{\pi}{2} \leq \alpha \leq \frac{\pi}{2} \\ \alpha - \pi, & \frac{\pi}{2} < \alpha < \pi \end{cases} = \begin{cases} \tan^{-1}x + \tan^{-1}y + \pi, & -\pi < \tan^{-1}x + \tan^{-1}y < -\frac{\pi}{2} \\ \tan^{-1}x + \tan^{-1}y, & -\frac{\pi}{2} \leq \tan^{-1}x + \tan^{-1}y \leq \frac{\pi}{2} \\ \tan^{-1}x + \tan^{-1}y - \pi, & \frac{\pi}{2} < \tan^{-1}x + \tan^{-1}y < \pi \end{cases}$$

Case-I :

$$-\pi < \tan^{-1}x + \tan^{-1}y < -\frac{\pi}{2} \Rightarrow x < 0, y < 0$$

$$\text{Also, } \tan^{-1}x < -\frac{\pi}{2} - \tan^{-1}y$$

$$\Rightarrow \tan^{-1}x < -\left(\frac{\pi}{2} - \tan^{-1}(-y)\right) \Rightarrow x < -\left(-\frac{1}{y}\right) \Rightarrow x < \frac{1}{y} \Rightarrow xy > 1$$

Case-II :

$$\frac{\pi}{2} < \tan^{-1}x + \tan^{-1}y < \pi \Rightarrow x, y > 0$$

$$\text{Also, } \tan^{-1}x > \frac{\pi}{2} - \tan^{-1}y \Rightarrow \tan^{-1}x > \tan^{-1}\frac{1}{y} \Rightarrow x > \frac{1}{y} \Rightarrow xy > 1$$

Case-III :

$$-\frac{\pi}{2} \leq \tan^{-1}x + \tan^{-1}y \leq \pi/2 \Rightarrow xy < 1$$

(2) $x > 0$ and $y > 0$, $\tan^{-1}x - \tan^{-1}y = \tan^{-1}\frac{x-y}{1+xy}$ (with no other restriction)

(Remember)

(i) $\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3 = \pi$

(ii) $\tan^{-1}1 + \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3} = \frac{\pi}{2}$

(iii) $\frac{\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3}{\cot^{-1}1 + \cot^{-1}2 + \cot^{-1}3} = 2$

Sol.(i) $\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3 = \tan^{-1}1 + \left(\pi + \tan^{-1}\frac{2+3}{1-2.3}\right)$
 $= \tan^{-1}1 + (\pi + \tan^{-1}(-1))$
 $= \frac{\pi}{4} + \pi - \frac{\pi}{4} = \pi$

$$\begin{aligned}
 \text{(ii)} \quad \tan^{-1} 1 + \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} &= \tan^{-1} 1 + \tan^{-1} \left(\frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \cdot \frac{1}{3}} \right) \\
 &= \tan^{-1} 1 + \tan^{-1} \left(\frac{5}{5} \right) = \tan^{-1} 1 + \tan^{-1} 1 = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2} \\
 \text{(iii)} \quad \frac{\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3}{\cot^{-1} 1 + \cot^{-1} 2 + \cot^{-1} 3} &= \frac{\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3}{\tan^{-1} 1 + \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3}} = \frac{\pi}{\left(\frac{\pi}{2} \right)} = 2
 \end{aligned}$$

Illustration :

If $\tan^{-1} 2 + \tan^{-1} 4 = \cot^{-1}(\lambda)$ then find λ .

$$\begin{aligned}
 \text{Sol.} \quad \tan^{-1} 2 + \tan^{-1} 4 &= \pi + \tan^{-1} \left(\frac{2+4}{1-2 \cdot 4} \right) = \pi + \tan^{-1} \left(\frac{6}{-7} \right) \\
 &= \pi - \tan^{-1} \frac{6}{7} = \pi - \cot^{-1} \frac{7}{6} = \cot^{-1} \left(-\frac{7}{6} \right) \Rightarrow \lambda = -\frac{7}{6} \text{ Ans.}
 \end{aligned}$$

Illustration :

If $\alpha = \tan^{-1} 5 - \tan^{-1} 3 + \tan^{-1} \frac{7}{9}$ and $\beta = \tan^{-1} \frac{2}{11} + \cot^{-1} \frac{24}{7} + \tan^{-1} \frac{1}{3}$, then
 (A) $\alpha = \beta$ (B) $\alpha > \beta$ (C) $\alpha < \beta$ (D) $\alpha + \beta = \pi/2$

$$\begin{aligned}
 \text{Sol.} \quad \alpha &= \tan^{-1} 5 - \tan^{-1} 3 + \tan^{-1} \frac{7}{9} \\
 &= \tan^{-1} \left(\frac{5-3}{1+5 \cdot 3} \right) + \tan^{-1} \frac{7}{9} = \tan^{-1} \frac{1}{8} + \tan^{-1} \frac{7}{9} \\
 &= \tan^{-1} \left(\frac{\frac{1}{8} + \frac{7}{9}}{1 - \frac{1}{8} \cdot \frac{7}{9}} \right) = \tan^{-1} \left(\frac{65}{65} \right) = \tan^{-1} 1 = \frac{\pi}{4} \\
 \beta &= \tan^{-1} \frac{2}{11} + \cot^{-1} \frac{24}{7} + \tan^{-1} \frac{1}{3} \\
 &= \tan^{-1} \frac{2}{11} + \tan^{-1} \frac{7}{24} + \tan^{-1} \frac{1}{3} = \tan^{-1} \frac{2}{11} + \tan^{-1} \left(\frac{\frac{7}{24} + \frac{1}{3}}{1 - \frac{7}{24} \cdot \frac{1}{3}} \right)
 \end{aligned}$$

$$\begin{aligned}
 &= \tan^{-1} \frac{2}{11} + \tan^{-1} \left(\frac{45}{65} \right) = \tan^{-1} \frac{2}{11} + \tan^{-1} \frac{9}{13} \\
 &= \tan^{-1} \left(\frac{\frac{2}{11} + \frac{9}{13}}{1 - \frac{2}{11} \cdot \frac{9}{13}} \right) = \tan^{-1} \left(\frac{125}{125} \right) = \tan^{-1} \frac{\pi}{4} \Rightarrow \alpha = \beta
 \end{aligned}$$

Illustration :

Find the value of $\cos^{-1} \sqrt{\frac{2}{3}} - \cos^{-1} \frac{\sqrt{6}+1}{2\sqrt{3}}$.

Sol. $\cos^{-1} \sqrt{\frac{2}{3}} - \cos^{-1} \frac{\sqrt{6}+1}{2\sqrt{3}} = \tan^{-1} \frac{1}{\sqrt{2}} - \tan^{-1} \left(\frac{\sqrt{3}-\sqrt{2}}{1+\sqrt{6}} \right) = \tan^{-1} \frac{1}{\sqrt{2}} - \tan^{-1} \left(\frac{\sqrt{3}-\sqrt{2}}{1+\sqrt{3}\cdot\sqrt{2}} \right)$

$$\begin{aligned}
 &= \tan^{-1} \frac{1}{\sqrt{2}} - (\tan^{-1} \sqrt{3} - \tan^{-1} \sqrt{2}) = \cot^{-1} \sqrt{2} - \tan^{-1} \sqrt{3} + \tan^{-1} \sqrt{2} \\
 &= \frac{\pi}{2} - \tan^{-1} \sqrt{3} = \cot^{-1} \sqrt{3} = \frac{\pi}{6} \text{ Ans.}
 \end{aligned}$$

Illustration :

Find the value of $\tan^{-1} \left(\frac{1}{2} \tan 2A \right) + \tan^{-1}(\cot A) + \tan^{-1}(\cot^3 A)$, for $0 < A < \frac{\pi}{4}$.

Sol. For $0 < A < \frac{\pi}{4}$, $\cot A > 1 \Rightarrow (\cot A)(\cot^3 A) > 1$

Then $\tan^{-1} \left(\frac{1}{2} \tan 2A \right) + \tan^{-1}(\cot A) + \tan^{-1}(\cot^3 A)$

$$\begin{aligned}
 &= \tan^{-1} \left(\frac{\tan A}{1 - \tan^2 A} \right) + \pi + \tan^{-1} \left(\frac{\cot A + \cot^3 A}{1 - \cot^4 A} \right) \\
 &= \tan^{-1} \left(\frac{\tan A}{1 - \tan^2 A} \right) + \pi + \tan^{-1} \left(\frac{\cot A}{1 - \cot^2 A} \right) \\
 &= \tan^{-1} \left(\frac{\tan A}{1 - \tan^2 A} \right) + \pi + \tan^{-1} \left(\frac{\tan A}{\tan^2 A - 1} \right) = \pi \text{ Ans.}
 \end{aligned}$$

Property-6 :

$$(I) \quad \sin^{-1}x + \sin^{-1}y = \begin{cases} \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2}) & \text{if } x \geq 0, y \geq 0 \text{ and } x^2 + y^2 \leq 1 \\ \pi - \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2}) & \text{if } x \geq 0, y \geq 0 \text{ and } x^2 + y^2 > 1 \end{cases}$$

note that $x^2 + y^2 \leq 1 \Rightarrow 0 \leq \sin^{-1}x + \sin^{-1}y \leq \frac{\pi}{2}$

Let $\sin^{-1}x = \alpha$ and $\sin^{-1}y = \beta$; $\alpha, \beta \in \left[0, \frac{\pi}{2}\right]$

now $x^2 + y^2 \leq 1$
 $\sin^2\alpha + \sin^2\beta \leq 1 \Rightarrow \sin^2\alpha \leq \cos^2\beta$

$$\sin^2\alpha \leq \sin^2\left(\frac{\pi}{2} - \beta\right) \Rightarrow \alpha \leq \frac{\pi}{2} - \beta \Rightarrow \alpha + \beta \leq \frac{\pi}{2}$$

$$0 \leq \sin^{-1}x + \sin^{-1}y \leq \frac{\pi}{2}$$

and $x^2 + y^2 > 1 \Rightarrow \frac{\pi}{2} < \sin^{-1}x + \sin^{-1}y < \pi$

This formula should normally be used in establishing the identities.

e.g. find whether $\sin^{-1}\frac{3}{5} + \sin^{-1}\frac{12}{13}$ is acute / obtuse will be unduly difficult using the above.

However if we convert it into $\tan^{-1}\frac{3}{4} + \tan^{-1}\frac{12}{5}$ it becomes simple.

$$(II) \quad \text{|| by we have } \sin^{-1}x - \sin^{-1}y = \sin^{-1}(x\sqrt{1-y^2} - y\sqrt{1-x^2}), x > 0, y > 0$$

and $\cos^{-1}x \pm \cos^{-1}y = \cos^{-1}(xy \mp \sqrt{1-x^2}\sqrt{1-y^2}), x > 0, y > 0, x < y$

Illustration :

Solve the equation $\sin^{-1}x + \sin^{-1}2x = \frac{\pi}{3}$.

Sol. $\sin^{-1}x + \sin^{-1}2x = \frac{\pi}{3}$.

$$\sin^{-1}2x = \sin^{-1}\frac{\sqrt{3}}{2} - \sin^{-1}x = \sin^{-1}\left[\frac{\sqrt{3}}{2}\sqrt{1-x^2} - x\sqrt{1-\frac{3}{4}}\right]$$

$$\Rightarrow 2x = \frac{\sqrt{3}}{2} \sqrt{1-x^2} - \frac{x}{2}$$

$$\Rightarrow \left(\frac{5x}{2}\right)^2 = \frac{3}{4}(1-x^2) \Rightarrow 28x^2 = 3$$

$$\Rightarrow x = \sqrt{\frac{3}{28}} = \frac{1}{2} \sqrt{\frac{3}{7}} \text{ Ans. } \left(\because x = -\frac{1}{2} \sqrt{\frac{3}{7}} \text{ makes L.H.S. of (1) negative} \right)$$

Illustration :

If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$ then value of $x^2 + y^2 + z^2 + 2xyz$ is equal to
(A) 1 (B) -1 (C) 0 (D) 3

Sol. $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi \Rightarrow \cos^{-1}x + \cos^{-1}y = \pi - \cos^{-1}z$

$$\Rightarrow \cos^{-1}\left(xy - \sqrt{1-x^2}\sqrt{1-y^2}\right) = \cos^{-1}(-z)$$

$$\Rightarrow xy - \sqrt{1-x^2}\sqrt{1-y^2} = -z$$

$$\Rightarrow xy + z = \sqrt{1-x^2}\sqrt{1-y^2}$$

squaring both sides

$$\Rightarrow x^2y^2 + z^2 + 2xyz = 1 - x^2 - y^2 + x^2y^2$$

$$\Rightarrow x^2 + y^2 + z^2 + 2xyz = 1.$$

Property-7 :

$$\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \tan^{-1}\left[\frac{x+y+z-xyz}{1-(xy+yz+zx)}\right]$$

where $x > 0, y > 0, z > 0$ and $xy + yz + zx < 1$ and $xy < 1, yz < 1, zx < 1$

Solution

$$\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \tan^{-1}\left(\frac{x+y}{1-xy}\right) + \tan^{-1}z$$

$$= \tan^{-1}\left(\frac{\frac{x+y}{1-xy} + z}{1 - \left(\frac{x+y}{1-xy}\right)z}\right) = \tan^{-1}\left(\frac{x+y+z-xyz}{1-xy-(x+y)z}\right) = \tan^{-1}\left(\frac{x+y+z-xyz}{1-(xy+yz+zx)}\right)$$

Practice Problem

- Q.1 Find the minimum value of $(\sec^{-1} x)^2 + (\operatorname{cosec}^{-1} x)^2$.
- Q.2 If two angles of a triangle are $\tan^{-1}(2)$ and $\tan^{-1}(3)$, then find the third angle.
- Q.3 Find x satisfying $\sin^{-1}\left(\frac{1}{\sqrt{5}}\right) + \cos^{-1}(x) = \frac{\pi}{4}$.
- Q.4 Value of $\sin^{-1}\left(\frac{3}{\sqrt{73}}\right) + \cos^{-1}\left(\frac{11}{\sqrt{146}}\right) + \cot^{-1}(\sqrt{3})$ is equal to
(A) π (B) $\pi/2$ (C) $5\pi/12$ (D) $\pi/3$
- Q.5 $\cos^{-1} x + \cos^{-1}\left(\frac{x}{2} + \frac{1}{2}\sqrt{3-3x^2}\right)$ is equal to
(A) $\frac{\pi}{3}$ for $x \in \left[\frac{1}{2}, 1\right]$ (B) $\frac{\pi}{3}$ for $x \in \left[0, \frac{1}{2}\right]$
(C) $2\cos^{-1} x - \cos^{-1} \frac{1}{2}$ for $x \in \left[\frac{1}{2}, 1\right]$ (D) $2\cos^{-1} x - \cos^{-1} \frac{1}{2}$ for $x \in \left[0, \frac{1}{2}\right]$

Answer key

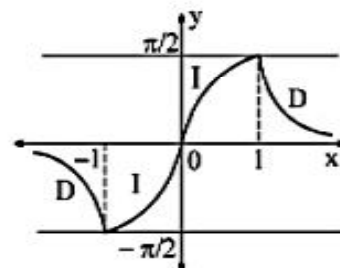
- Q.1 $\frac{\pi^2}{8}$ Q.2 $\frac{\pi}{4}$ Q.3 $x = \frac{3}{\sqrt{10}}$ Q.4 C Q.5 A, D

SIMPLIFICATION & TRANSFORMATION OF INVERSE FUNCTIONS BY ELEMENTARY SUBSTITUTION AND THEIR GRAPHS :

$$(I) \quad \sin^{-1} \frac{2x}{1+x^2} = \begin{cases} 2 \tan^{-1} x & -1 \leq x \leq 1 \\ \pi - 2 \tan^{-1} x & \text{if } x \geq 1 \\ -\pi - 2 \tan^{-1} x & x \leq -1 \end{cases}$$

Proof:

$$\text{Let } x = \tan \theta, \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \Rightarrow \theta = \tan^{-1} x$$



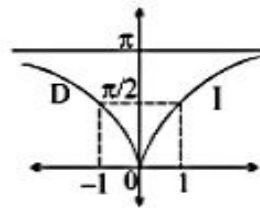
Now, $\sin^{-1}\left(\frac{2x}{1+x^2}\right) = \sin^{-1}\left(\frac{2 \tan \theta}{1 + \tan^2 \theta}\right) = \sin^{-1}(\sin 2\theta) = \sin^{-1}(\sin \alpha)$, where $\alpha \in (-\pi, \pi)$

$$\sin^{-1}\left(\frac{2x}{1+x^2}\right) = \sin^{-1}(\sin \alpha)$$

$$= \begin{cases} -\alpha - \pi, & -\pi < \alpha < -\frac{\pi}{2} \\ \alpha, & -\frac{\pi}{2} \leq \alpha \leq \frac{\pi}{2} \\ -\alpha + \pi, & \frac{\pi}{2} < \alpha < \pi \end{cases} = \begin{cases} -2 \tan^{-1} x - \pi, & -\pi < 2 \tan^{-1} x < -\frac{\pi}{2} \\ 2 \tan^{-1} x, & -\frac{\pi}{2} \leq 2 \tan^{-1} x \leq \frac{\pi}{2} \\ -2 \tan^{-1} x + \pi, & \frac{\pi}{2} < 2 \tan^{-1} x < \pi \end{cases}$$

$$= \begin{cases} -2 \tan^{-1} x - \pi, & -\frac{\pi}{2} < \tan^{-1} x < -\frac{\pi}{4} \\ 2 \tan^{-1} x, & -\frac{\pi}{4} \leq \tan^{-1} x \leq \frac{\pi}{4} \\ -2 \tan^{-1} x + \pi, & \frac{\pi}{4} < \tan^{-1} x < \frac{\pi}{2} \end{cases} = \begin{cases} -2 \tan^{-1} x - \pi, & x < -1 \\ 2 \tan^{-1} x, & -1 \leq x \leq 1 \\ -2 \tan^{-1} x + \pi, & x > 1 \end{cases}$$

(2) $\cos^{-1} \frac{1-x^2}{1+x^2} = \begin{cases} 2 \tan^{-1} x & x \geq 0 \\ -2 \tan^{-1} x & x < 0 \end{cases}$



Proof:

Let $x = \tan \theta$, $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \Rightarrow \theta = \tan^{-1} x$

Now, $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \cos^{-1}\left(\frac{1-\tan^2 \theta}{1+\tan^2 \theta}\right) = \cos^{-1}(\cos 2\theta) = \cos^{-1}(\cos \alpha)$, where $\alpha \in (-\pi, \pi)$

$$\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \cos^{-1}(\cos \alpha)$$

$$= \begin{cases} -\alpha, & -\pi < \alpha < 0 \\ \alpha, & 0 \leq \alpha < \pi \end{cases} = \begin{cases} -2 \tan^{-1} x, & -\pi < 2 \tan^{-1} x < 0 \\ 2 \tan^{-1} x, & 0 \leq 2 \tan^{-1} x < \pi \end{cases}$$

$$= \begin{cases} -2 \tan^{-1} x, & -\frac{\pi}{2} < \tan^{-1} x < 0 \\ 2 \tan^{-1} x, & 0 \leq \tan^{-1} x < \frac{\pi}{2} \end{cases} = \begin{cases} -2 \tan^{-1} x, & x < 0 \\ 2 \tan^{-1} x, & x \geq 0 \end{cases}$$

$$(3) \quad \tan^{-1} \frac{2x}{1-x^2} = \begin{cases} \pi + 2 \tan^{-1} x & x < -1 \\ 2 \tan^{-1} x & -1 < x < 1 \\ 2 \tan^{-1} x - \pi & x > 1 \end{cases}$$

Proof:

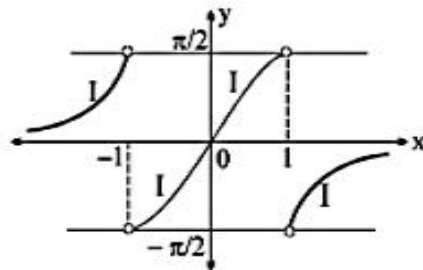
$$\text{Let } x = \tan \theta, \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \Rightarrow \theta = \tan^{-1} x$$

$$\text{Now, } \tan^{-1} \left(\frac{2x}{1-x^2} \right) = \tan^{-1} \left(\frac{2 \tan \theta}{1 - \tan^2 \theta} \right) = \tan^{-1}(\tan 2\theta) = \tan^{-1}(\tan \alpha), \text{ where } \alpha \in (-\pi, \pi)$$

$$\tan^{-1} \left(\frac{2x}{1-x^2} \right) = \tan^{-1}(\tan \alpha)$$

$$= \begin{cases} \alpha + \pi, & -\pi < \alpha < -\frac{\pi}{2} \\ \alpha, & -\frac{\pi}{2} \leq \alpha \leq \frac{\pi}{2} \\ \alpha - \pi, & \frac{\pi}{2} < \alpha < \pi \end{cases} = \begin{cases} 2 \tan^{-1} x + \pi, & -\pi < 2 \tan^{-1} x < -\frac{\pi}{2} \\ 2 \tan^{-1} x, & -\frac{\pi}{2} \leq 2 \tan^{-1} x \leq \frac{\pi}{2} \\ 2 \tan^{-1} x - \pi, & \frac{\pi}{2} < 2 \tan^{-1} x < \pi \end{cases}$$

$$= \begin{cases} 2 \tan^{-1} x + \pi, & -\frac{\pi}{2} < \tan^{-1} x < -\frac{\pi}{4} \\ 2 \tan^{-1} x, & -\frac{\pi}{4} \leq \tan^{-1} x \leq \frac{\pi}{4} \\ 2 \tan^{-1} x - \pi, & \frac{\pi}{4} < \tan^{-1} x < \frac{\pi}{2} \end{cases} = \begin{cases} \pi + 2 \tan^{-1} x & x < -1 \\ 2 \tan^{-1} x & -1 \leq x \leq 1 \\ 2 \tan^{-1} x - \pi & x > 1 \end{cases}$$



Highlights :-

- (a) $f(x) = \sin^{-1} \frac{2x}{1+x^2} + 2 \tan^{-1} x = \pi$ if $x \geq 1$
- (b) $f(x) = \sin^{-1} \frac{2x}{1+x^2} + 2 \tan^{-1} x = -\pi$ if $x \leq -1$

Illustration :

If $f(x) = \sin^{-1} \frac{2x}{1+x^2} + 2 \tan^{-1} x$ then find

- (a) $f(100)$ (b) $\cos(f(-10))$

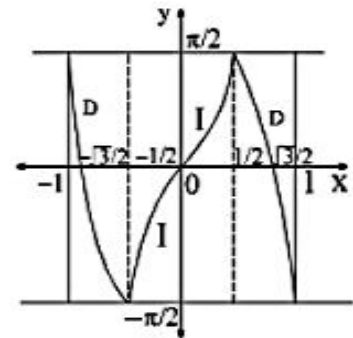
Sol. We know that

$$f(x) = \sin^{-1} \frac{2x}{1+x^2} + 2 \tan^{-1} x = \pi \text{ if } x \geq 1$$

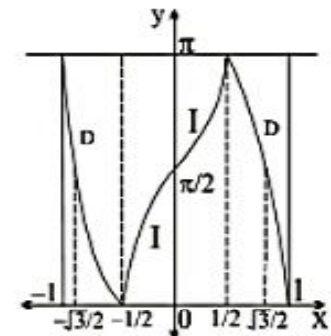
$$f(x) = \sin^{-1} \frac{2x}{1+x^2} + 2 \tan^{-1} x = -\pi \text{ if } x \leq -1$$

- $\Rightarrow$ (a) $f(100) = \pi$
(b) $\cos(f(-10)) = \cos(-\pi) = -1$

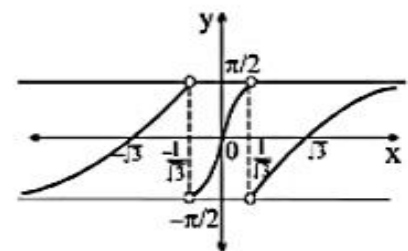
$$(4) \quad \sin^{-1}(3x - 4x^3) = \begin{cases} -(\pi + 3\sin^{-1} x) & \text{if } -1 \leq x \leq -1/2 \\ 3\sin^{-1} x & \text{if } -1/2 \leq x \leq 1/2 \\ \pi - 3\sin^{-1} x & \text{if } 1/2 \leq x \leq 1 \end{cases};$$



$$(5) \quad \cos^{-1}(4x^3 - 3x) = \begin{cases} 3\cos^{-1} x - 2\pi & \text{if } -1 \leq x \leq -1/2 \\ 2\pi - 3\cos^{-1} x & \text{if } -1/2 \leq x \leq 1/2 \\ 3\cos^{-1} x & \text{if } 1/2 \leq x \leq 1 \end{cases};$$



$$(6) \quad \tan^{-1} \frac{3x - x^3}{1 - 3x^2} = \begin{cases} 3\tan^{-1} x & \text{if } -\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}} \\ -\pi + 3\tan^{-1} x & \text{if } x > \frac{1}{\sqrt{3}} \\ \pi + 3\tan^{-1} x & \text{if } x < -\frac{1}{\sqrt{3}} \end{cases}$$



* (4, 5, 6 to be proved similarly as 1, 2, 3)

(C) IDENTITIES INVOLVING INVERSE TRIGONOMETRIC FUNCTIONS:

$$(I) \quad 2\tan^{-1}\left(\tan\left(\frac{\pi}{4} - \alpha\right)\tan\frac{\beta}{2}\right) = \cos^{-1}\left(\frac{\sin 2\alpha + \cos \beta}{1 + \sin 2\alpha \cos \beta}\right)$$

Proof: Let $x = \tan\left(\frac{\pi}{4} - \alpha\right)\tan\frac{\beta}{2}$

$$x = \frac{1 - \tan \alpha}{1 + \tan \alpha} \tan \frac{\beta}{2} \Rightarrow x = \left(\frac{\cos \alpha - \sin \alpha}{\cos \alpha + \sin \alpha}\right) \frac{\sin \frac{\beta}{2}}{\cos \frac{\beta}{2}}$$

$$x^2 = \left(\frac{1 - \sin 2\alpha}{1 + \sin 2\alpha}\right) \frac{\sin^2 \frac{\beta}{2}}{\cos^2 \frac{\beta}{2}}$$

$$x^2 = \frac{(1 - \sin 2\alpha)(1 - \cos \beta)}{(1 + \sin 2\alpha)(1 + \cos \beta)} = \frac{1 - \sin 2\alpha - \cos \beta + \sin 2\alpha \cdot \cos \beta}{1 + \sin 2\alpha + \cos \beta + \sin 2\alpha \cdot \cos \beta}$$

$$\frac{x^2 - 1}{x^2 + 1} = \frac{-(\sin 2\alpha + \cos \beta)}{(1 + \sin 2\alpha \cdot \cos \beta)}$$

(By applying componendo and dividendo)

$$\frac{1 - x^2}{1 + x^2} = \frac{\sin 2\alpha + \cos \beta}{1 + \sin 2\alpha \cdot \cos \beta}$$

We know that

$$2 \tan^{-1} x = \cos^{-1} \left(\frac{1 - x^2}{1 + x^2} \right)$$

$$\Rightarrow 2 \tan^{-1} \left(\tan \left(\frac{\pi}{4} - \alpha \right) \tan \frac{\beta}{2} \right) = \cos^{-1} \left(\frac{\sin 2\alpha + \cos \beta}{1 + \sin 2\alpha \cos \beta} \right)$$

(II) $\tan^{-1} x = 2 \tan^{-1} [\operatorname{cosec}(\tan^{-1} x) - \tan(\cot^{-1} x)] \quad (x \neq 0)$

Sol. R.H.S. $2 \tan^{-1} [\operatorname{cosec}(\tan^{-1} x) - \tan(\cot^{-1} x)]$

$$= 2 \tan^{-1} \left[\operatorname{cosec}(\tan^{-1} x) - \tan \left(\frac{\pi}{2} - \tan^{-1} x \right) \right]$$

$$= 2 \tan^{-1} [\operatorname{cosec}(\tan^{-1} x) - \cot \tan^{-1} x]$$

$$\text{let } \tan^{-1} x = \theta$$

$$\Rightarrow 2 \tan^{-1} [\operatorname{cosec} \theta - \cot \theta]$$

$$= 2 \tan^{-1} \left[\frac{1 - \cos \theta}{\sin \theta} \right] = 2 \tan^{-1} \left[\frac{2 \sin^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cdot \cos \frac{\theta}{2}} \right] = 2 \tan^{-1} \tan \frac{\theta}{2} = 2 \left(\frac{\theta}{2} \right)$$

$$= \theta = \tan^{-1} x$$

L.H.S.

(D) EQUATIONS INVOLVING INVERSE TRIGONOMETRIC FUNCTIONS :

Illustration :

Solve the equation $\tan^{-1} \frac{x-1}{x+2} + \tan^{-1} \frac{x+1}{x+2} = \frac{\pi}{4}$.

Sol. $\tan^{-1} \frac{x-1}{x+2} + \tan^{-1} \frac{x+1}{x+2} = \frac{\pi}{4}$

$$\Rightarrow \tan^{-1} \left[\frac{\frac{x-1}{x+2} + \frac{x+1}{x+2}}{1 - \left(\frac{x-1}{x+2} \right) \left(\frac{x+1}{x+2} \right)} \right] = \frac{\pi}{4} \Rightarrow \left[\frac{2x(x+2)}{x^2 + 4 + 4x - x^2 + 1} \right] = \tan \frac{\pi}{4}$$

$$\Rightarrow \frac{2x(x+2)}{4x+5} = 1 \Rightarrow 2x^2 + 4x = 4x + 5 \Rightarrow x = \pm \sqrt{\frac{5}{2}}$$

But for $x = -\sqrt{\frac{5}{2}}$, L.H.S. is negative. Hence $x = \sqrt{\frac{5}{2}}$.

Illustration :

Find the x satisfying the equation $2\cot^{-1}2 - \cos^{-1}\frac{4}{5} = \operatorname{cosec}^{-1}x$.

Sol. $2\cot^{-1}2 - \cos^{-1}\frac{4}{5} = 2\tan^{-1}\frac{1}{2} - \cos^{-1}\frac{4}{5}$

$$= \tan^{-1} \frac{2\left(\frac{1}{2}\right)}{1 - \left(\frac{1}{2}\right)^2} - \tan^{-1} \frac{3}{4} = \tan^{-1} \left(\frac{4}{3}\right) - \tan^{-1} \frac{3}{4}$$

$$= \tan^{-1} \left(\frac{\frac{4}{3} - \frac{3}{4}}{1 + \frac{4}{3} \cdot \frac{3}{4}} \right) = \tan^{-1} \left(\frac{7}{24} \right) = \operatorname{cosec}^{-1} \left(\frac{25}{7} \right)$$

$$= \operatorname{cosec}^{-1}x = \operatorname{cosec}^{-1} \left(\frac{25}{7} \right)$$

$$x = \frac{25}{7} \text{ Ans.}$$

Illustration :

Find the x satisfying the equation $\sin[2 \cos^{-1}\{\cot(2\tan^{-1}x)\}] = 0$

Sol. $\sin[2 \cos^{-1}\{\cot(2\tan^{-1}x)\}] = 0$

$$\Rightarrow 2 \cos^{-1}\{\cot(2\tan^{-1}x)\} = n\pi, n \in I$$

$$\Rightarrow \cos^{-1}\{\cot(2\tan^{-1}x)\} = \frac{n\pi}{2} \Rightarrow \cot(2\tan^{-1}x) = \cos \frac{n\pi}{2}$$

$\cos \frac{n\pi}{2}$ can take the values $\pm 1, 0$ for $n \in I$.

Case-I : When $\cos \frac{n\pi}{2} = \pm 1$

$$\Rightarrow \cot(2\tan^{-1}x) = \pm 1 \Rightarrow 2\tan^{-1}x = n\pi \pm \frac{\pi}{4}$$

$$\Rightarrow \tan^{-1}x = \frac{n\pi}{2} \pm \frac{\pi}{8} \Rightarrow x = \tan\left(\frac{n\pi}{2} \pm \frac{\pi}{8}\right) \Rightarrow x = \pm \tan \frac{\pi}{8}, \pm \cot \frac{\pi}{8}$$

$$x = \pm(\sqrt{2} - 1), \pm(\sqrt{2} + 1)$$

$$\Rightarrow x = \pm(\sqrt{2} \pm 1)$$

Case-II : When $\cos \frac{n\pi}{2} = 0$

$$\Rightarrow \cot(2\tan^{-1}x) = 0 \Rightarrow 2\tan^{-1}x = n\pi + \frac{\pi}{2}$$

$$\Rightarrow \tan^{-1}x = \frac{n\pi}{2} + \frac{\pi}{4} \Rightarrow x = \tan\left(\frac{n\pi}{2} + \frac{\pi}{4}\right) \Rightarrow x = \tan\frac{\pi}{4}, -\cot\frac{\pi}{4}$$

$$x = \pm 1$$

Case-I and Case-II

$$x = \pm 1, \pm(\sqrt{2} \pm 1) \text{ Ans.}$$

(E) SIMULTANEOUS EQUATIONS AND INEQUALITIES INVOLVING I.T.F. :

Illustration :

Find the x satisfying the inequality $\cos^{-1}x > \cos^{-1}x^2$.

Sol. $\cos^{-1}x > \cos^{-1}x^2$

$$\Rightarrow x^2 - x > 0 \Rightarrow x(x-1) > 0 \Rightarrow x \in (-\infty, 0) \cup (1, \infty)$$

$$\because \cos^{-1}x \text{ defined for } x \in [-1, 1]$$

$$\Rightarrow x \in [-1, 0) \text{ Ans.}$$

Illustration :

Solve the inequality satisfying

$$\text{arc tan}^2 x - 3 \text{ arc tan} x + 2 > 0$$

where $[]$ denotes the greatest integer function.

Sol. $(\tan^{-1}x)^2 - 3\tan^{-1}x + 2 > 0$

$$\Rightarrow (\tan^{-1}x - 1)(\tan^{-1}x - 2) > 0$$

$$\because \tan^{-1}x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\Rightarrow (\tan^{-1}x - 2) \text{ is always negative}$$

$$\Rightarrow (\tan^{-1}x - 1)(\tan^{-1}x - 2) > 0 \text{ holds true only when } \tan^{-1}x - 1 < 0$$

$$\Rightarrow \tan^{-1}x < 1 \Rightarrow x < \tan 1$$

$$\Rightarrow x \in (-\infty, \tan 1) \text{ Ans.}$$

(F) SUMMATION OF SERIES :

Illustration :

Prove that

$$\tan^{-1} \frac{2}{2+1^2+1^4} + \tan^{-1} \frac{4}{2+2^2+2^4} + \tan^{-1} \frac{6}{2+3^2+3^4} + \dots \text{upto } n \text{ terms} = \tan^{-1} (n(n+1)+1) - \frac{\pi}{4}$$

Sol. $T_r = \tan^{-1} \left(\frac{2r}{2+r^2+r^4} \right) = \tan^{-1} \left(\frac{2r}{1+(r^2+1)^2-r^2} \right)$

$$= \tan^{-1} \left[\frac{(r^2+r+1)-(r^2-r+1)}{1+(r^2+r+1)(r^2-r+1)} \right]$$

$$= \tan^{-1} (r^2+r+1) - \tan^{-1} (r^2-r+1)$$

$$S_n = \sum_{r=1}^n T_r = (\tan^{-1} 3 - \tan^{-1} 1) + (\tan^{-1} 7 - \tan^{-1} 3) + (\tan^{-1} 13 - \tan^{-1} 7)$$

$$+ \dots + (\tan^{-1} (n^2+n+1) - \tan^{-1} (n^2-n+1))$$

$$S_n = \tan^{-1} (n^2+n+1) - \tan^{-1} 1 = \tan^{-1} (n(n+1)+1) - \frac{\pi}{4}$$

Illustration :

Prove that

$$\tan^{-1} \frac{x}{1+(1 \times 2)x^2} + \tan^{-1} \frac{x}{1+(2 \times 3)x^2} + \dots + \tan^{-1} \frac{x}{1+n(n+1)x^2} = \tan^{-1} (n+1)x - \tan^{-1} x.$$

Sol. $T_r = \tan^{-1} \frac{x}{1+r(r+1)x^2} = \tan^{-1} \left(\frac{(r+1)x - rx}{1+r(r+1)x^2} \right) = \tan^{-1} (r+1)x - \tan^{-1} rx$

$$S_n = \sum_{r=1}^n T_r = (\tan^{-1} 2x - \tan^{-1} x) + (\tan^{-1} 3x - \tan^{-1} 2x) + \dots + (\tan^{-1} (n+1)x - \tan^{-1} nx)$$

$$S_n = \tan^{-1} (n+1)x - \tan^{-1} x$$

Illustration :

The value of $\operatorname{cosec}^{-1} \sqrt{5} + \operatorname{cosec}^{-1} \sqrt{65} + \operatorname{cosec}^{-1} \sqrt{325} + \dots \infty$ is equal to

- (A) 0 (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) π

Sol. $\operatorname{cosec}^{-1} \sqrt{5} + \operatorname{cosec}^{-1} \sqrt{65} + \operatorname{cosec}^{-1} \sqrt{325} + \dots \infty$.

$$= \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{8} + \tan^{-1} \frac{1}{18} + \dots \infty$$

$$T_r = \tan^{-1} \frac{1}{2r^2} = \tan^{-1} \frac{2}{4r^2} = \tan^{-1} \left(\frac{(2r+1) - (2r-1)}{1 + (2r-1)(2r+1)} \right)$$

$$T_r = \tan^{-1}(2r+1) - \tan^{-1}(2r-1)$$

$$S_n = \sum_{r=1}^n T_r = (\tan^{-1} 3 - \tan^{-1} 1) + (\tan^{-1} 5 - \tan^{-1} 3) + \dots + (\tan^{-1}(2n+1) - \tan^{-1}(2n-1))$$

$$= \tan^{-1}(2n+1) - \tan^{-1} 1$$

when $n \rightarrow \infty$

$$S_\infty = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4} \text{ Ans.}$$

Illustration :

The value of $S = \sum_{n=1}^{\infty} \tan^{-1} \left(\frac{4n}{n^4 - 2n^2 + 2} \right)$ is equal to

- (A) $3 \tan^{-1} 1$ (B) $\cot^{-1} \operatorname{sgn}(x^2 + 1)$ (C) $\frac{\pi}{4}$ (D) $\tan^{-1} 2 + \tan^{-1} 3$

Sol. $S = \sum_{n=1}^{\infty} \tan^{-1} \left(\frac{4n}{n^4 - 2n^2 + 2} \right) = \sum_{n=1}^{\infty} \tan^{-1} \left(\frac{4n}{1 + (n^2 - 1)^2} \right)$

$$= \sum_{n=1}^{\infty} \tan^{-1} \left(\frac{(n+1)^2 - (n-1)^2}{1 + (n+1)^2(n-1)^2} \right) = \sum_{n=1}^{\infty} [\tan^{-1}(n+1)^2 - \tan^{-1}(n-1)^2]$$

$$= (\tan^{-1} 2^2 - \tan^{-1} 0) + (\tan^{-1} 3^2 - \tan^{-1} 1^2) + \dots + (\tan^{-1} n^2 - \tan^{-1} (n-2)^2) + (\tan^{-1} (n+1)^2 - \tan^{-1} (n-1)^2)$$

$$\Rightarrow S = \tan^{-1} (n+1)^2 + \tan^{-1} n^2 - (\tan^{-1} 1^2 + \tan^{-1} 0)$$

$$= \left(\frac{\pi}{2} + \frac{\pi}{2} \right) - \left(\frac{\pi}{4} + 0 \right) (\because n \rightarrow \infty) = \frac{3\pi}{4} \text{ Ans.}$$

Practice Problem

- Q.1 If $(x-1)(x^2+1) > 0$, then find the value of $\sin\left(\frac{1}{2}\tan^{-1}\frac{2x}{1-x^2} - \tan^{-1}x\right)$
- Q.2 If $\cos^{-1}\frac{6x}{1+9x^2} = -\frac{\pi}{2} + 2\tan^{-1}3x$, then find the value of x .
- Q.3 Solve the equation $[\sin^{-1}x] > [\cos^{-1}x]$
where $[]$ denotes the greatest integer function.
- Q.4 Value of x satisfying the equation $2\cot^{-1}2 + \cos^{-1}(3/5) = \operatorname{cosec}^{-1}x$ is
(A) ϕ (B) 1 (C) $25/7$ (D) $25/24$
- Q.5 Sum of the series $\cot^{-1}(2a^{-1}+a) + \cot^{-1}(2a^{-1}+3a) + \cot^{-1}(2a^{-1}+6a) + \cot^{-1}(2a^{-1}+10a) + \dots$ to ∞ is equal to ($a > 0$)
(A) $\tan^{-1}\left(\frac{a}{2}\right)$ (B) $\cot^{-1}\left(\frac{a}{2}\right)$ (C) $a + \frac{1}{a}$ (D) $2a$
- Q.6 If $\cos^{-1}x - \sin^{-1}x = \cos^{-1}x\sqrt{3}$ then value(s) of x satisfying
(A) 0 (B) $\frac{1}{2}$ (C) 1 (D) $-\frac{1}{2}$

Answer key

- Q.1 -1 Q.2 $x \in \left(\frac{1}{3}, \infty\right)$ Q.3 $x \in (\sin 1, 1)$ Q.4 A
- Q.5 B Q.6 A, B, D

Solved Examples

Q.1 Domain of definition of the function $f(x) = \sqrt{3 \cos^{-1}(4x) - \pi}$ is equal to

- (A) $\left[-\frac{1}{4}, \frac{1}{8}\right]$ (B) $\left[\frac{1}{8}, 1\right]$ (C) $\left[\frac{1}{8}, \frac{1}{4}\right]$ (D) $\left[-1, \frac{1}{8}\right]$

Sol. For domain of $f(x) = \sqrt{3 \cos^{-1}(4x) - \pi}$, we must have

$$\cos^{-1} 4x \geq \frac{\pi}{3} \Rightarrow 4x \leq \frac{1}{2} \Rightarrow x \leq \frac{1}{8} \quad \dots\dots(1)$$

$$\text{Also } -1 \leq 4x \leq 1 \Rightarrow \frac{-1}{4} \leq x \leq \frac{1}{4} \quad \dots\dots(2)$$

$$\therefore \text{ From (1) and (2), we get } x \in \left[-\frac{1}{4}, \frac{1}{8}\right]$$

Q.2 If $a \sin^{-1} x - b \cos^{-1} x = c$, then the value of $a \sin^{-1} x + b \cos^{-1} x$ (whenever exists) is equal to

- (A) 0 (B) $\frac{\pi ab + c(b-a)}{a+b}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi ab + c(a-b)}{a+b}$

Sol. We have $b \sin^{-1} x + b \cos^{-1} x = \frac{b\pi}{2} \quad \dots\dots(1)$

$$\text{and } a \sin^{-1} x - b \cos^{-1} x = c \quad \dots\dots(2) \quad (\text{given})$$

$$\therefore \text{ On adding (1) and (2), we get } (a+b) \sin^{-1} x = \frac{b\pi}{2} + c$$

$$\Rightarrow \sin^{-1} x = \frac{\frac{b\pi}{2} + c}{a+b}. \quad \text{Similarly } \cos^{-1} x = \frac{\frac{a\pi}{2} - c}{a+b}$$

$$\text{Hence } (a \sin^{-1} x + b \cos^{-1} x) = \frac{\pi ab + c(a-b)}{a+b}$$

Q.3 If $0 < \cos^{-1} x < 1$ and $1 + \sin(\cos^{-1} x) + \sin^2(\cos^{-1} x) + \sin^3(\cos^{-1} x) + \dots\dots \infty = 2$, then x equals

- (A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2\sqrt{3}}$

Sol. We have $1 + \sin(\cos^{-1} x) + \sin^2(\cos^{-1} x) + \dots\dots \infty = 2$

$$\Rightarrow \frac{1}{1 - \sin(\cos^{-1} x)} = 2 \Rightarrow \frac{1}{2} = 1 - \sin(\cos^{-1} x) \Rightarrow \sin(\cos^{-1} x) = \frac{1}{2}$$

$$\Rightarrow \cos^{-1} x = \frac{\pi}{6} \Rightarrow x = \frac{\sqrt{3}}{2} \quad \text{Ans.}$$

Q.4 If $\tan^{-1} \left(x + \frac{3}{x} \right) - \tan^{-1} \left(x - \frac{3}{x} \right) = \tan^{-1} \frac{6}{x}$, then the value of $5x^8 - 4x^4 + 7$ equals

- (A) 397 (B) 393 (C) 376 (D) 379

Sol. We have $\tan^{-1} \left(x + \frac{3}{x} \right) - \tan^{-1} \left(x - \frac{3}{x} \right) = \tan^{-1} \frac{6}{x}$

$$\Rightarrow \tan^{-1} \left(\frac{\left(x + \frac{3}{x} \right) - \left(x - \frac{3}{x} \right)}{1 + \left(x + \frac{3}{x} \right) \left(x - \frac{3}{x} \right)} \right) = \tan^{-1} \frac{6}{x} \Rightarrow x^2 - \frac{9}{x^2} = 0 \Rightarrow x^4 = 9$$

Hence $(5x^8 - 4x^4 + 7) = 5(81) - 4(9) + 7 = 405 - 36 + 7 = 412 - 36 = 376$.

Q.5 The value of $\tan^{-1} \frac{4}{7} + \tan^{-1} \frac{4}{19} + \tan^{-1} \frac{4}{39} + \tan^{-1} \frac{4}{67} + \dots \infty$ equals

- (A) $\tan^{-1} 1 + \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3}$ (B) $\tan^{-1} 1 + \cot^{-1} 3$
(C) $\cot^{-1} 1 + \cot^{-1} \frac{1}{2} + \cot^{-1} \frac{1}{3}$ (D) $\cot^{-1} 1 + \tan^{-1} 3$

Sol. Let $S = 7 + 19 + 39 + 67 + \dots + T_n$
 $S = 0 + 7 + 19 + 39 + \dots + T_{n-1} + T_n$
 (Subtracting) $\quad - \quad - \quad - \quad - \quad - \quad - \quad -$

$$T_n = 7 + 12 + 20 + 28 + \dots + (T_n - T_{n-1})$$

$$= 7 + \frac{(n-1)}{2} [24 + 8(n-2)] = 4n^2 + 3$$

$$\therefore T_n' = \tan^{-1} \frac{4}{4n^2 + 3} = \tan^{-1} \frac{1}{n^2 + \frac{3}{4}} = \tan^{-1} \frac{1}{1 + \left(n^2 - \frac{1}{4} \right)}$$

$$= \tan^{-1} \left[\frac{\left(n + \frac{1}{2} \right) - \left(n - \frac{1}{2} \right)}{1 + \left(n + \frac{1}{2} \right) \left(n - \frac{1}{2} \right)} \right] = \tan^{-1} \left(n + \frac{1}{2} \right) - \tan^{-1} \left(n - \frac{1}{2} \right)$$

$$\text{Hence } S_{\infty} = \sum_{n=1}^{\infty} T_n' = \frac{\pi}{2} - \tan^{-1} \frac{1}{2} = \tan^{-1} 1 + \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} - \tan^{-1} \frac{1}{2} = \tan^{-1} 1 + \cot^{-1} 3$$

Q.6 Number of values of x satisfying the equation

$$\tan^{-1} \left(x - \frac{x^3}{4} + \frac{x^5}{16} - \dots \right) + \cot^{-1} \left(x + \frac{x^2}{2} + \frac{x^3}{4} + \dots \right) = \frac{\pi}{2} \text{ for } 0 < |x| < 2, \text{ is}$$

- (A) 0 (B) 1 (C) 2 (D) 3

Sol. We must have $x - \frac{x^3}{4} + \frac{x^5}{16} - \dots = x + \frac{x^2}{2} + \frac{x^3}{4} + \dots$

$$\Rightarrow \frac{x}{1 + \frac{x^2}{4}} = \frac{x}{1 - \frac{x}{2}} \Rightarrow \frac{4x}{4 + x^2} = \frac{2x}{2 - x} \Rightarrow 2x^2(x + 2) = 0$$

$\therefore x = 0, -2$ (As $0 < |x| < 2$)

Clearly, no value of x satisfies given equation.

Q.7 Number of integral ordered pairs (x, y) satisfying the equation $\arctan \frac{1}{x} + \arctan \frac{1}{y} = \arctan \frac{1}{10}$, is

- (A) 1 (B) 2 (C) 3 (D) 4

Sol. Since $\tan(\arctan a) = a \forall a \in \mathbb{R} = a \forall a \in \mathbb{R}$,

Take $\tan$ both side

$$\tan \left(\arctan \frac{1}{x} + \arctan \frac{1}{y} \right) = \tan \left(\arctan \frac{1}{10} \right)$$

$$\Rightarrow \frac{\frac{1}{x} + \frac{1}{y}}{1 - \frac{1}{xy}} = \frac{1}{10} \Rightarrow (x - 10)(y - 10) = 101$$

The following four ordered pair of integer numbers are solutions of this equation :

$(11, 111); (111, 11), (9, -91) \Rightarrow$ ordered pairs **Ans.**

Q.8 For $n \in \mathbb{N}$, if $\tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{4} + \tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{n} = \frac{\pi}{4}$ then n is equal to

- (A) 43 (B) 47 (C) 49 (D) 51

Sol. We have, $\tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{4} = \tan^{-1} \left(\frac{\frac{1}{3} + \frac{1}{4}}{1 - \frac{1}{12}} \right) = \tan^{-1} \left(\frac{7}{11} \right)$

$$\text{Again, } \tan^{-1} \frac{7}{11} + \tan^{-1} \frac{1}{5} = \tan^{-1} \left(\frac{\frac{7}{11} + \frac{1}{5}}{1 - \frac{7}{55}} \right) = \tan^{-1} \left(\frac{46}{48} \right) = \tan^{-1} \frac{23}{24}$$

$$\therefore \tan^{-1} \frac{1}{n} = \tan^{-1} 1 - \tan^{-1} \frac{23}{24} = \tan^{-1} \left(\frac{1 - \frac{23}{24}}{1 + \frac{23}{24}} \right) = \tan^{-1} \left(\frac{1}{47} \right) \Rightarrow n = 47 \text{ Ans.}$$

Q.9 If $0 < \cos^{-1}x < 1$ and $1 + \cos^{-1}x + (\cos^{-1}x)^2 + \dots \infty = 2$ then x is equal to

- (A) $\frac{\pi}{4}$ (B) $\cos \frac{1}{2}$ (C) $\cos \frac{1}{\sqrt{2}}$ (D) $\frac{\pi}{6}$

Sol. We have

$1 + \cos^{-1}x + (\cos^{-1}x)^2 + \dots \infty = 2$, (which is an infinite geometric progression)

$$\Rightarrow \frac{1}{1 - \cos^{-1}x} = 2 \Rightarrow \cos^{-1}x = \frac{1}{2} \Rightarrow x = \cos \frac{1}{2} \quad \text{Ans.}$$

Q.10 The domain of definition of function $f(x) = \sqrt{\cos^{-1}x - 2\sin^{-1}x}$ is equal to

- (A) $\left[-\frac{1}{2}, \frac{1}{2}\right]$ (B) $\left[\frac{1}{2}, 1\right]$ (C) $\left[-1, \frac{1}{2}\right]$ (D) $\left[-1, \frac{\sqrt{3}}{2}\right]$

Sol. We have $f(x) = \sqrt{\cos^{-1}x - 2\sin^{-1}x}$

Clearly, for domain of $f(x)$, $\cos^{-1}x - 2\sin^{-1}x \geq 0$

$$\Rightarrow \frac{\pi}{2} \geq 3\sin^{-1}x \Rightarrow \sin^{-1}x \leq \frac{\pi}{6} \Rightarrow x \leq \frac{1}{2}$$

$$\text{So, } D_f = \left[-1, \frac{1}{2}\right] \quad \text{Ans.}$$

Paragraph for question nos. 11 to 13

In $\triangle ABC$, if $\angle B = \sec^{-1}\left(\frac{5}{4}\right) + \operatorname{cosec}^{-1}\sqrt{5}$, $\angle C = \operatorname{cosec}^{-1}\left(\frac{25}{7}\right) + \cot^{-1}\left(\frac{9}{13}\right)$ and $c = 3$.

(All symbols used have their usual meaning in a triangle.)

Q.11 $\tan A, \tan B, \tan C$ are in

- (A) A.P. (B) G.P. (C) H.P. (D) neither A.P, G.P. nor H.P.

Q.12 The distance between orthocentre and centroid of triangle with sides $a^2, b^{\frac{4}{3}}$ and c is equal to

- (A) $\frac{5}{2}$ (B) $\frac{5}{3}$ (C) $\frac{10}{3}$ (D) $\frac{7}{2}$

Q.13 Which of the following is rational with respect to $\triangle ABC$?

- (A) r_1 (B) r_2 (C) r_3 (D) Δ

$$\text{Sol. } \angle B = \sec^{-1}\left(\frac{5}{4}\right) + \operatorname{cosec}^{-1}\sqrt{5} = \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{1}{2}\right) = \tan^{-1} \frac{\frac{3}{4} + \frac{1}{2}}{1 - \frac{3}{4} \cdot \frac{1}{2}} = \tan^{-1} 2$$

$$\angle C = \operatorname{cosec}^{-1}\left(\frac{25}{7}\right) + \cot^{-1}\left(\frac{9}{13}\right) = \tan^{-1}\left(\frac{7}{24}\right) + \tan^{-1}\left(\frac{13}{9}\right) = \tan^{-1} \frac{\frac{7}{24} + \frac{13}{9}}{1 - \frac{7}{24} \cdot \frac{13}{9}} = \tan^{-1} 3$$

$$\therefore \angle A = \pi - \angle B - \angle C = \pi - \tan^{-1} 2 - \tan^{-1} 3 = \tan^{-1} 1$$

$$\therefore \sin A = \frac{1}{\sqrt{2}}, \sin B = \frac{2}{\sqrt{5}} \text{ and } \sin C = \frac{3}{\sqrt{10}}$$

$$\therefore \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \Rightarrow \sqrt{2}a = \frac{\sqrt{5}b}{2} = \frac{c\sqrt{10}}{3}$$

Hence $a = \sqrt{5}$ and $b = 2\sqrt{2}$, $c = 3$

(i) $\tan A = 1$, $\tan B = 2$, $\tan C = 3$ are in A.P.

(ii) The triangle with sides a^2 , $b^{\frac{4}{3}}$ and c will have side-length 5, 4 and 3 respectively

$$\therefore \text{Distance between orthocentre and centroid} = \frac{2}{3} (\text{circumradius}) = \frac{\text{hypotenuse}}{3} = \frac{5}{3} \text{ Ans.}$$

(iii) Area of $\triangle ABC$, $\Delta = \frac{1}{2} ab \sin C = \frac{1}{2} (\sqrt{5}) (2\sqrt{2}) \left(\frac{3}{\sqrt{10}} \right) = 3$

$$\text{Also } s = \frac{1}{2}(a+b+c) = \frac{1}{2}(\sqrt{5} + 2\sqrt{2} + 3)$$

$$\therefore s-a = \frac{1}{2}(-\sqrt{5} + 2\sqrt{2} + 3), s-b = \frac{1}{2}(\sqrt{5} - 2\sqrt{2} + 3) \text{ and } s-c = \frac{1}{2}(\sqrt{5} + 2\sqrt{2} - 3)$$

$\therefore \Delta$ is rational

$\therefore$ Each of values $\frac{\Delta}{s-a}$, $\frac{\Delta}{s-b}$ and $\frac{\Delta}{s-c}$ i.e. r_1, r_2 and r_3 (respectively) will be irrational.

Q.14 If $(\sin^{-1} x)^2 + (\sin^{-1} y)^2 + (\sin^{-1} z)^2 = \frac{3\pi^2}{4}$, then the value of $(x-y+z)$ can be

(A) 1 (B) -1 (C) 3 (D) -3

Sol. As $-\frac{\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2} \quad \forall -1 \leq x \leq 1$

$$\therefore (\sin^{-1} x)^2 + (\sin^{-1} y)^2 + (\sin^{-1} z)^2 = \frac{\pi^2}{4} + \frac{\pi^2}{4} + \frac{\pi^2}{4} \text{ is possible if } x, y, z \in \{-1, 1\}$$

$\therefore$ Possible values of $x-y+z$ from the ordered triplet (x, y, z) are as follows :

(x, y, z)	$(x-y+z)$
$(-1, -1, -1)$	-1
$(-1, 1, 1)$	-1
$(1, -1, 1)$	3
$(1, 1, -1)$	-1
$(1, 1, 1)$	1
$(1, -1, -1)$	-1
$(-1, 1, -1)$	-3
$(-1, -1, 1)$	1

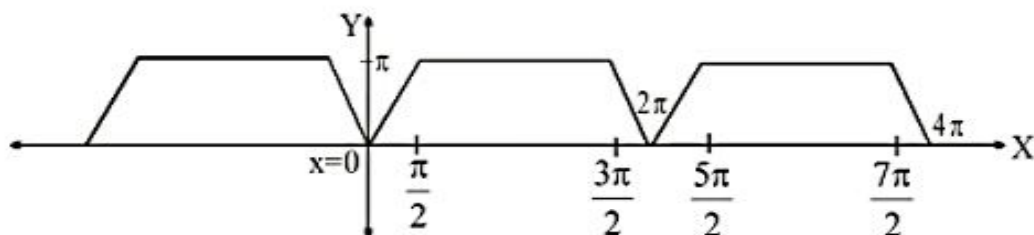
Hence set of values of $(x-y+z)$ is $\{\pm 1, \pm 3\}$

Q.15 Let $f(x) = \sin^{-1} |\sin x| + \cos^{-1}(\cos x)$. Which of the following statement(s) is/are **TRUE** ?

- (A) $f(f(3)) = \pi$ (B) $f(x)$ is periodic with fundamental period 2π .
(C) $f(x)$ is neither even nor odd. (D) Range of $f(x)$ is $[0, 2\pi]$

Sol.
$$f(x) = \begin{cases} 2x & ; 0 \leq x \leq \frac{\pi}{2} \\ \pi & ; \frac{\pi}{2} < x \leq \frac{3\pi}{2} \\ 4\pi - 2x & ; \frac{3\pi}{2} < x \leq 2\pi \end{cases}$$

Clearly $f(x)$ is periodic function with period 2π . The graph of $f(x)$ is shown below.



Q.16 If $f(x) = \sin^{-1} x \cdot \cos^{-1} x \cdot \tan^{-1} x \cdot \cot^{-1} x \cdot \sec^{-1} x \cdot \operatorname{cosec}^{-1} x$, then which of the following statement(s) hold(s) good ?

- (A) The graph of $y = f(x)$ does not lie above x axis.
(B) The non-negative difference between maximum and minimum value of the function $y = f(x)$ is $\frac{3\pi^6}{64}$.
(C) The function $y = f(x)$ is not injective.
(D) Number of non-negative integers in the domain of $f(x)$ is two.

Sol. Domain of $\sin^{-1} x$ and $\cos^{-1} x$, each is $[-1, 1]$ and that of $\sec^{-1} x$ and $\operatorname{cosec}^{-1} x$, each is $(-\infty, -1] \cup [1, \infty)$
 $\therefore$ Domain of $f(x)$ must be $\{-1, 1\}$ $\therefore$ Range of $f(x)$ will be $\{f(-1), f(1)\}$
where $f(-1) = \sin^{-1}(-1) \cdot \cos^{-1}(-1) \cdot \tan^{-1}(-1) \cdot \cot^{-1}(-1) \cdot \sec^{-1}(-1) \cdot \operatorname{cosec}^{-1}(-1)$

$$= \left(-\frac{\pi}{2}\right) \cdot (\pi) \cdot \left(-\frac{\pi}{4}\right) \cdot \left(\frac{3\pi}{4}\right) \cdot (\pi) \cdot \left(-\frac{\pi}{2}\right) = \frac{-3\pi^6}{64} \text{ and } f(1) = 0 \text{ \{as } \cos^{-1} 1 = 0\}}$$

(i) Thus, the graph of $f(x)$ is a two point graph which doesn't lie above x - axis.

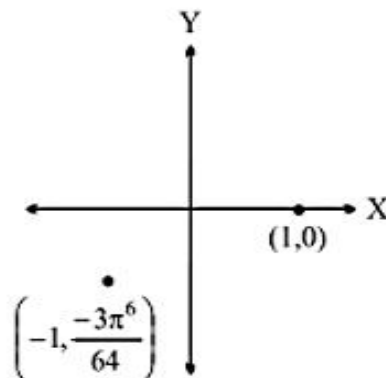
(ii) $f(x)_{\max} = 0$ and $f(x)_{\min} = \frac{-3\pi^6}{64}$

Hence $|f(x)_{\max} - f(x)_{\min}| = \frac{3\pi^6}{64}$

(iii) $f(x)$ is one-one hence injective.

(iv) Domain is $\{-1, 1\}$

$\therefore$ Number of non-negative integers in the domain of $f(x)$ is one.



- Q.17 Consider $f(x) = \tan^{-1} \left(\frac{(\sqrt{12}-2)x^2}{x^4+2x^2+3} \right)$ and m and M are respectively minimum and maximum values of $f(x)$ and $x = a$ ($a > 0$) is the point in the domain of $f(x)$ where $f(x)$ attains its maximum value.

Column I	Column II
(A) If $\sin^{-1} 2\sqrt{x} = 3 \tan^{-1}(\tan(m+M))$ then $8x$ equals	(P) 0
(B) If $\cos^{-1} x + \cos^{-1} y = 3 \left\{ \tan^{-1} \left(\tan \frac{7M}{2} \right) + \tan^{-1} \left(m + \tan \frac{3\pi}{8} \right) \right\}$ then $(x+y)$ equals	(Q) 2
(C) The value of $\tan \left(\sec^{-1} \left(\frac{2}{a^2} \right) + M \right)$ equals	(R) -2
(D) If α and β are roots of the equation $x^2 - (\tan(3 \sin^{-1}(\sin M)))x + a^4 = 0$, then $\alpha\beta - (\alpha + \beta)$ equals	(S) 1
	(T) -1

Sol. We have $f(x) = \tan^{-1} \left(\frac{2(\sqrt{3}-1)}{x^2 + \frac{3}{x^2} + 2} \right)$

As $x^2 + \frac{3}{x^2} \geq 2\sqrt{3}$ (Using A.M. - G.M. inequality)

$\Rightarrow x^2 + \frac{3}{x^2} + 2 \geq 2 + 2\sqrt{3}$

$\therefore f(x)|_{\max} = \tan^{-1} \left(\frac{2(\sqrt{3}-1)}{2(\sqrt{3}+1)} \right) = \frac{\pi}{12} = M$, which occurs at $x^2 = \frac{3}{x^2} \Rightarrow x = 3^{\frac{1}{4}} = a$

$f(x)|_{\min} = 0 = m$, which occurs at $x = 0$

(A) $\sin^{-1}(2\sqrt{x}) = 3 \tan^{-1} \left(\tan \frac{\pi}{12} \right) = \frac{\pi}{4}$

$2\sqrt{x} = \frac{1}{\sqrt{2}} \Rightarrow 8x = 1$

(B) $\cos^{-1} x + \cos^{-1} y = 3 \left[\tan^{-1} \left(\tan \frac{7\pi}{24} \right) + \tan^{-1} \left(0 + \tan \frac{3\pi}{8} \right) \right] = 3 \left[\frac{7\pi}{24} + \frac{3\pi}{8} \right] = 3 \left(\frac{16\pi}{24} \right) = 2\pi$

$\therefore x = y = -1 \Rightarrow x + y = -2$

(C) $\tan \left(\sec^{-1} \left(\frac{2}{\sqrt{3}} \right) + \frac{\pi}{12} \right) = \tan \left(\frac{\pi}{6} + \frac{\pi}{12} \right) = 1$

(D) $x^2 - \tan\left(3\sin^{-1}\left(\sin\frac{\pi}{12}\right)\right)x + 3 = 0$

$$x^2 - x + 3 = 0 \begin{cases} \alpha \\ \beta \end{cases}$$

$$\therefore \alpha + \beta = 1$$

$$\therefore \alpha\beta = 3$$

$$\text{Hence } \alpha\beta - (\alpha + \beta) = 2$$

Q.18 Let $\alpha = 3\cos^{-1}\left(\frac{5}{\sqrt{28}}\right) + 3\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$ and $\beta = 4\sin^{-1}\left(\frac{7\sqrt{2}}{10}\right) - 4\tan^{-1}\left(\frac{3}{4}\right)$

then which of the following does not hold(s) good?

(A) $\alpha < \pi$ but $\beta > \pi$.

(B) $\alpha > \pi$ but $\beta < \pi$.

(C) Both α and β are equal.

(D) $\cos(\alpha + \beta) = 0$.

Sol. $\alpha = 3\tan^{-1}\left(\frac{\sqrt{3}}{5}\right) + 3\tan^{-1}\left(\frac{\sqrt{3}}{2}\right) = 3\left[\tan^{-1}\frac{\frac{\sqrt{3}}{5} + \frac{\sqrt{3}}{2}}{1 - \frac{3}{10}}\right] = 3\tan^{-1}\left(\frac{7\sqrt{3}}{7}\right) = \pi$.

$$\beta = 4\left[\tan^{-1}7 - \tan^{-1}\frac{3}{4}\right] = 4\left[\tan^{-1}\frac{7 - \frac{3}{4}}{1 + \frac{21}{4}}\right] = 4\tan^{-1}\left(\frac{25}{25}\right) = \pi.$$

Q.19 If range of the function $f(x) = \sin^{-1}x + 2\tan^{-1}x + x^2 + 4x + 1$ is $[p, q]$ then find the value of $(p + q)$.

Sol. We have $f(x) = \sin^{-1}x + 2\tan^{-1}x + x^2 + 4x + 1$

Clearly domain of $f(x)$ is $[-1, 1]$.

Also $f(x)$ is increasing function in its domain.

$$\therefore p = f_{\min.}(x) = f(-1) = -\frac{\pi}{2} + 2\left(\frac{-\pi}{4}\right) + 1 - 4 + 1 = -\pi - 2.$$

$$q = f_{\max.}(x) = f(1) = \frac{\pi}{2} + 2\left(\frac{\pi}{4}\right) + 1 + 4 + 1 = \pi + 6.$$

$\therefore$ Range of $f(x)$ is $[-\pi - 2, \pi + 6]$

Hence $(p + q) = 4$

Note : Vertex of $y = x^2 + 4x + 1$ is at $x = -2$ and hence in the domain $(x^2 + 4x + 1)$ is increasing.

Q.20 Let α, β, γ and δ be the roots of equation $x^4 - 3x^3 + 5x^2 - 7x + 9 = 0$. If the value of $|\tan(\tan^{-1}\alpha + \tan^{-1}\beta + \tan^{-1}\gamma + \tan^{-1}\delta)| = \frac{a}{b}$ where a and b are coprime to each other, then find the value of $(a^b + b^a + a^a + b^b + ab)$.

Sol. From given equation, we have

$$S_1 = \Sigma\alpha = 3, \quad S_2 = \Sigma\alpha\beta = 5$$

$$S_3 = \Sigma\alpha\beta\gamma = 7 \text{ and } S_4 = \alpha\beta\gamma\delta = 9$$

$$\text{Let } \tan^{-1}\alpha = A, \tan^{-1}\beta = B, \tan^{-1}\gamma = C \text{ \& } \tan^{-1}\delta = D$$

Now $|\tan(\tan^{-1}\alpha + \tan^{-1}\beta + \tan^{-1}\gamma + \tan^{-1}\delta)|$

$$= |\tan(A + B + C + D)| = \left| \frac{S_1 - S_3}{1 - S_2 + S_4} \right| = \left| \frac{3 - 7}{1 - 5 + 9} \right| = \frac{4}{5} = \frac{a}{b}$$

Hence $a = 4$ and $b = 5$

So $(a^b + b^a + a^a + b^b + ab) = 4^5 + 5^4 + 4^4 + 5^5 + 4.5 = 1024 + 625 + 256 + 3125 + 20 = 5050$ Ans.

Q.21 How many terms of the sequence $\cot^{-1} 3, \cot^{-1} 7, \cot^{-1} 13, \cot^{-1} 21, \dots$ must be taken to have their sum equal to $\frac{1}{2} \cos^{-1} \left(\frac{24}{145} \right)$.

Sol. $T_1 = \tan^{-1} \frac{1}{3} = \tan^{-1} 2 - \tan^{-1} 1$; $T_2 = \tan^{-1} \frac{1}{7} = \tan^{-1} 3 - \tan^{-1} 2$; $T_3 = \tan^{-1} \frac{1}{13} = \tan^{-1} 4 - \tan^{-1} 3$

Clearly $T_n = \tan^{-1}(n+1) - \tan^{-1}(n)$

$$\text{Hence } S_n = \tan^{-1}(n+1) - \tan^{-1} 1 = \tan^{-1} \left(\frac{n+1-1}{1+(n+1) \cdot 1} \right) = \left(\tan^{-1} \frac{n}{n+2} \right) = \frac{1}{2} \cos^{-1} \left(\frac{24}{145} \right)$$

$$\Rightarrow 2 \left(\tan^{-1} \frac{n}{n+2} \right) = \cos^{-1} \left(\frac{24}{145} \right) \quad \left(\text{Using } 2 \tan^{-1} x = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) \forall x \geq 0 \right)$$

$$\Rightarrow \cos^{-1} \left(\frac{2(n+1)}{n^2 + 2n + 2} \right) = \cos^{-1} \left(\frac{24}{145} \right) \Rightarrow \left(\frac{2(n+1)}{n^2 + 2n + 2} \right) = \left(\frac{24}{145} \right)$$

$$\Rightarrow 12(n+1)^2 - 144(n+1) - (n+1) + 12 = 0 = ((n+1) - 12)(12(n+1) - 1) = 0$$

$$\therefore n+1 = 12, \frac{1}{12} \quad \therefore n = 11, \frac{-11}{12} \quad \because n \in \mathbb{N} \quad \therefore n \neq \frac{-11}{12}$$

Hence, $n = 11$ Ans.

Q.22 If the area enclosed by the curves $f(x) = \cos^{-1}(\cos x)$ and $g(x) = \sin^{-1}(\cos x)$ in $x \in \left[\frac{9\pi}{4}, \frac{15\pi}{4} \right]$ is $\frac{a\pi^2}{b}$ (where a and b are coprime), then find $(a+b)$.

Sol. We have $g(x) = \sin^{-1}(\cos x) = \frac{\pi}{2} - \cos^{-1}(\cos x)$

Both the curves bound the regions of same area

in $\left[\frac{\pi}{4}, \frac{7\pi}{4} \right], \left[\frac{9\pi}{4}, \frac{15\pi}{4} \right]$ and so on

$$\therefore \text{Required area} = \text{area of shaded square} = \frac{9\pi^2}{8} = \frac{a\pi^2}{b}$$

$\therefore a = 9$ and $b = 8$

Hence $a + b = 17$ Ans.

